SUSTech Southern University of Science and Technology

SME206，信号与系统

Lab2 – 线性时不变系统

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Outline

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- Discrete-Time LTI Systems
- Continuous-Time LTI Systems
- Showcase: Echo Cancellation



Outline

- **Discrete-Time LTI Systems**
- Continuous-Time LTI Systems
- Showcase: Echo Cancellation

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Background: DT LTI System

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$$\delta[n-k] \rightarrow \boxed{S} \rightarrow h_k[n] \quad (k = \cdots, -1, 0, 1, \cdots)$$

S is linear

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k] \rightarrow \boxed{S} \rightarrow y[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h_k[n]$$

S is time-invariant

$$h_k[n] = h[n-k]$$

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\delta[n-k] \rightarrow \boxed{S} \rightarrow y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$

Tutorial: conv

```
>> help conv
 conv Convolution and polynomial multiplication.
    C = conv(A, B) convolves vectors A and B.  The resulting vector is
    length MAX([LENGTH(A)+LENGTH(B)-1,LENGTH(A),LENGTH(B)]). If A and B are
    vectors of polynomial coefficients, convolving them is equivalent to
    multiplying the two polynomials.

    C = conv(A, B, SHAPE) returns a subsection of the convolution with size
    specified by SHAPE:
      'full'  - (default) returns the full convolution,
      'same'  - returns the central part of the convolution
                that is the same size as A.
      'valid' - returns only those parts of the convolution
                that are computed without the zero-padded edges.
                LENGTH(C)is MAX(LENGTH(A)-MAX(0,LENGTH(B)-1),0).

    Class support for inputs A,B:
      float: double, single
```

Highlight

- **输入：两个向量 A, B**
- **返回：向量 C**
  - **长度：length(A) + length(B)-1**
    - A[k] 与 B[n-k] 交叠部分
  - **问题：时间维度对应是什么?**
    - e.g, C(1) 对应 n 是什么?

Tutorial: conv / showcase

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k]$$

$$x[n] = \begin{cases} 1, & 1 \le n \le 5 \\ 0, & otherwise \end{cases}$$

$$h[n] = \begin{cases} 1, & 2 \le n \le 6 \\ 0, & otherwise \end{cases}$$

```
nx = 1:5
x = [1, 1, 1, 1, 1];

nh = 2:6
h = [1, 1, 1, 1, 1];

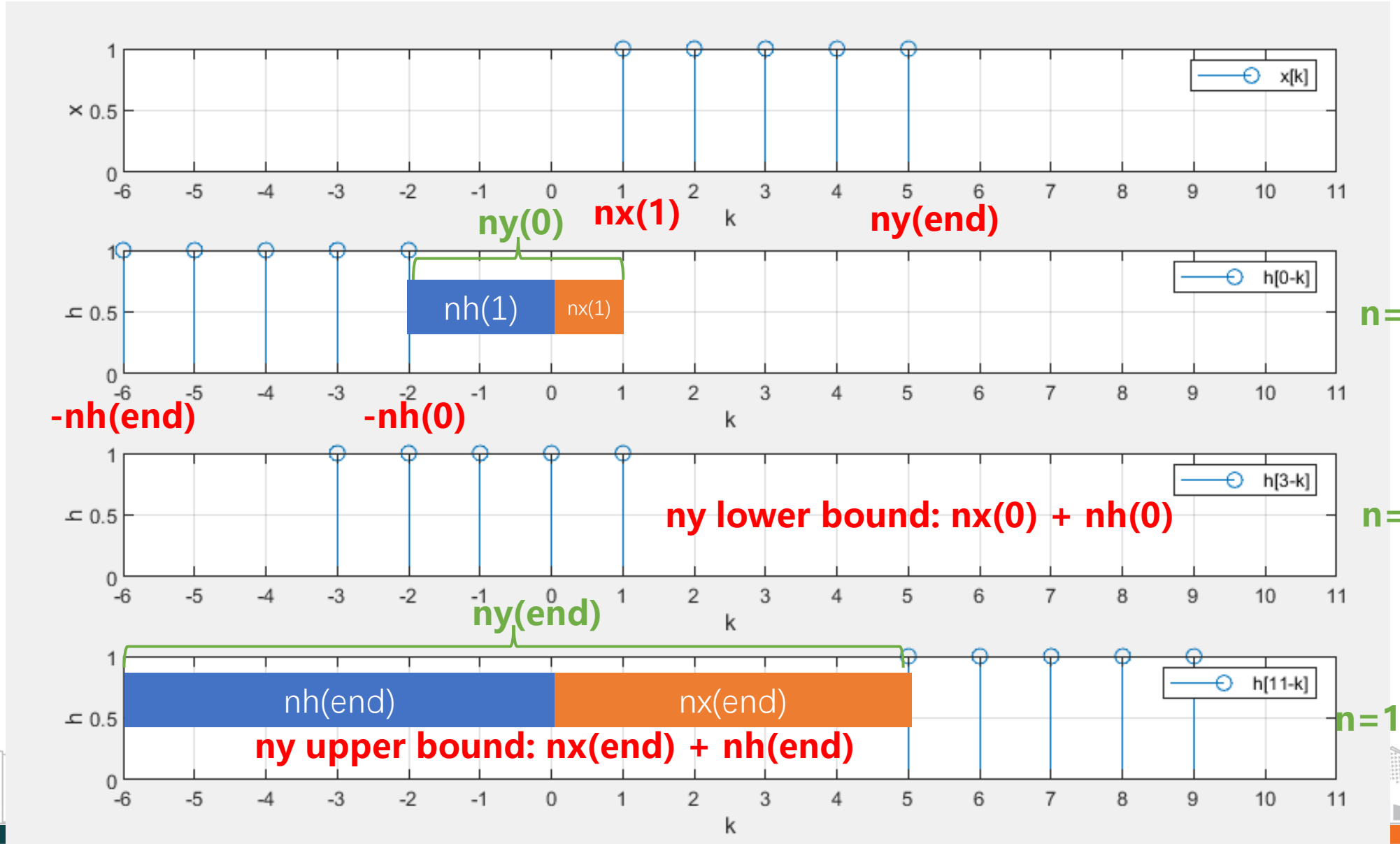
y = conv(x, h);
```

=> 问题：y 对应时间 ny 是什么?

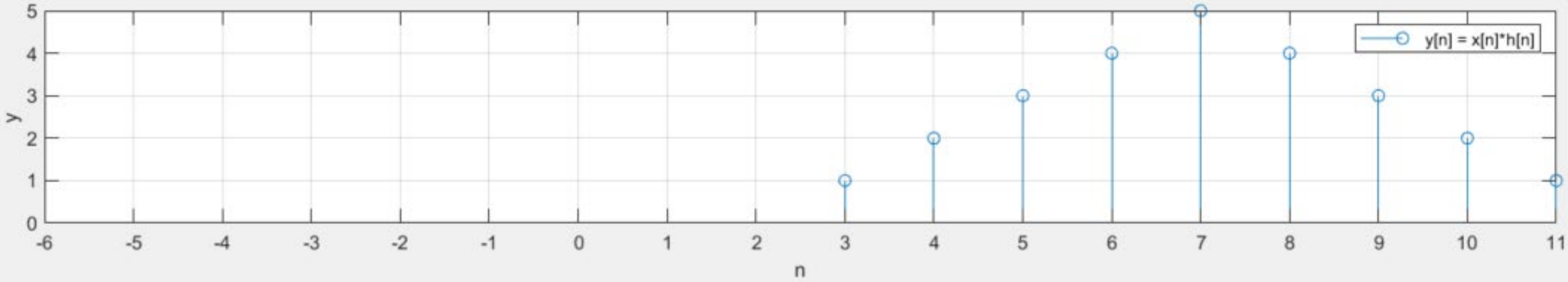
Tutorial: conv / showcase

• 问题：y对应时间ny是什么?

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Tutorial: conv / showcase




```
ny = (nx(1) + nh(1)) : (nx(end) + nh(end));
y = conv(x, h);
```

Background: Linear Constant-Coefficient Difference Equation

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$$\sum_{k=0}^{N} a_k\, y[n-k] = \sum_{k=0}^{M} b_k\, x[n-k]$$

generally, a0 ≠ 0

$$y[n] = \sum_{k=0}^{M} \frac{b_k}{a_0}\, x[n-k] - \sum_{k=1}^{N} \frac{a_k}{a_0}\, y[n-k]$$

when $a_0 \neq 0,\ a_k = 0\ (k = 1, 2, \dots N)$

$$y[n] = \sum_{k=0}^{M} \frac{b_k}{a_0}\, x[n-k] = x[n] * h[n]$$

$$h[n] = \begin{cases} \frac{b_k}{a_0}, & 0 \le k \le M \\ 0, & otherwise \end{cases}$$

$\vec{A} = [a_0, a_1, a_2, \dots, a_N]$

$\vec{B} = [b_0, b_1, b_2, \dots, b_M]$

Linear constant-coefficient difference equation + The condition of initial rest = Causal LTI system

e.g,
if x[n] = 0 for n < n0,
then y[n] = 0 for all n < n0 as well

Connect to: convolution sum

Tutorial: filter

Highlight

- 输入：
  - 线性常系数方程系数 B, A 向量
  - 系统输入 X
- 返回：Y
  - 长度：与 X 相同
  - 注意：存在截断？

```
>> help filter
 filter One-dimensional digital filter.
   Y = filter(B,A,X) filters the data in vector X with the
   filter described by vectors A and B to create the filtered
   data Y.  The filter is a "Direct Form II Transposed"
   implementation of the standard difference equation:

   a(1)*y(n) = b(1)*x(n) + b(2)*x(n-1) + ... + b(nb+1)*x(n-nb)
                         - a(2)*y(n-1) - ... - a(na+1)*y(n-na)

   If a(1) is not equal to 1, filter normalizes the filter
   coefficients by a(1).

   filter always operates along the first non-singleton dimension,
   namely dimension 1 for column vectors and non-trivial matrices,
   and dimension 2 for row vectors.

   [Y,Zf] = filter(B,A,X,Zi) gives access to initial and final
   conditions, Zi and Zf, of the delays.  Zi is a vector of length
   MAX(LENGTH(A),LENGTH(B))-1, or an array with the leading dimension
   of size MAX(LENGTH(A),LENGTH(B))-1 and with remaining dimensions
   matching those of X.

   filter(B,A,X,[],DIM) or filter(B,A,X,Zi,DIM) operates along the
   dimension DIM.

   Tip:  If you have the Signal Processing Toolbox, you can design a
   filter, D, using DESIGNFILT.  Then you can use Y = filter(D,X) to
   filter your data.
```

Tutorial: filter / showcase

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From Lab Book, 2.2) a, b, c, d

(a). Define coefficient vectors `a1` and `b1` to describe the causal LTI system specified by $y[n] = 0.5x[n] + x[n-1] + 2x[n-2]$.

(b). Define coefficient vectors `a2` and `b2` to describe the causal LTI system specified by $y[n] = 0.8y[n-1] + 2x[n]$.

(c). Define coefficient vectors `a3` and `b3` to describe the causal LTI system specified by $y[n] - 0.8y[n-1] = 2x[n-1]$.

```
a1 = [1];
b1 = [0.5, 1, 2];
```

```
a2 = [1, -0.8];
b2 = [2];
```

```
a3 = [1, -0.8];
b3 = [0, 2];
```

Tutorial: filter / showcase

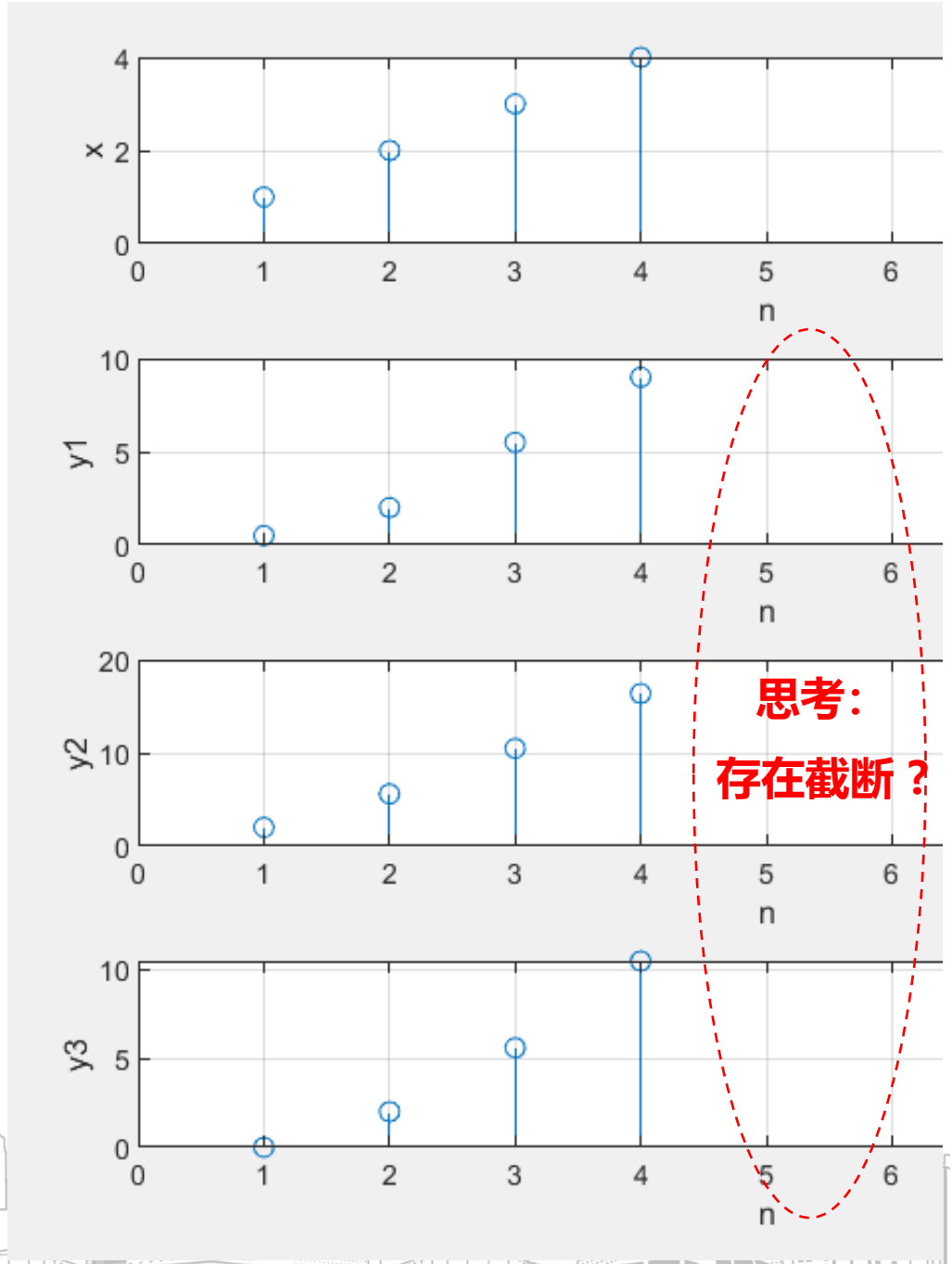
From Lab Book, 2.2) a, b, c, d

```
n = 1:4;
x = n;

y1 = filter(b1, a1, x);

y2 = filter(b2, a2, x);

y3 = filter(b3, a3, x);
```



Tutorial: filte

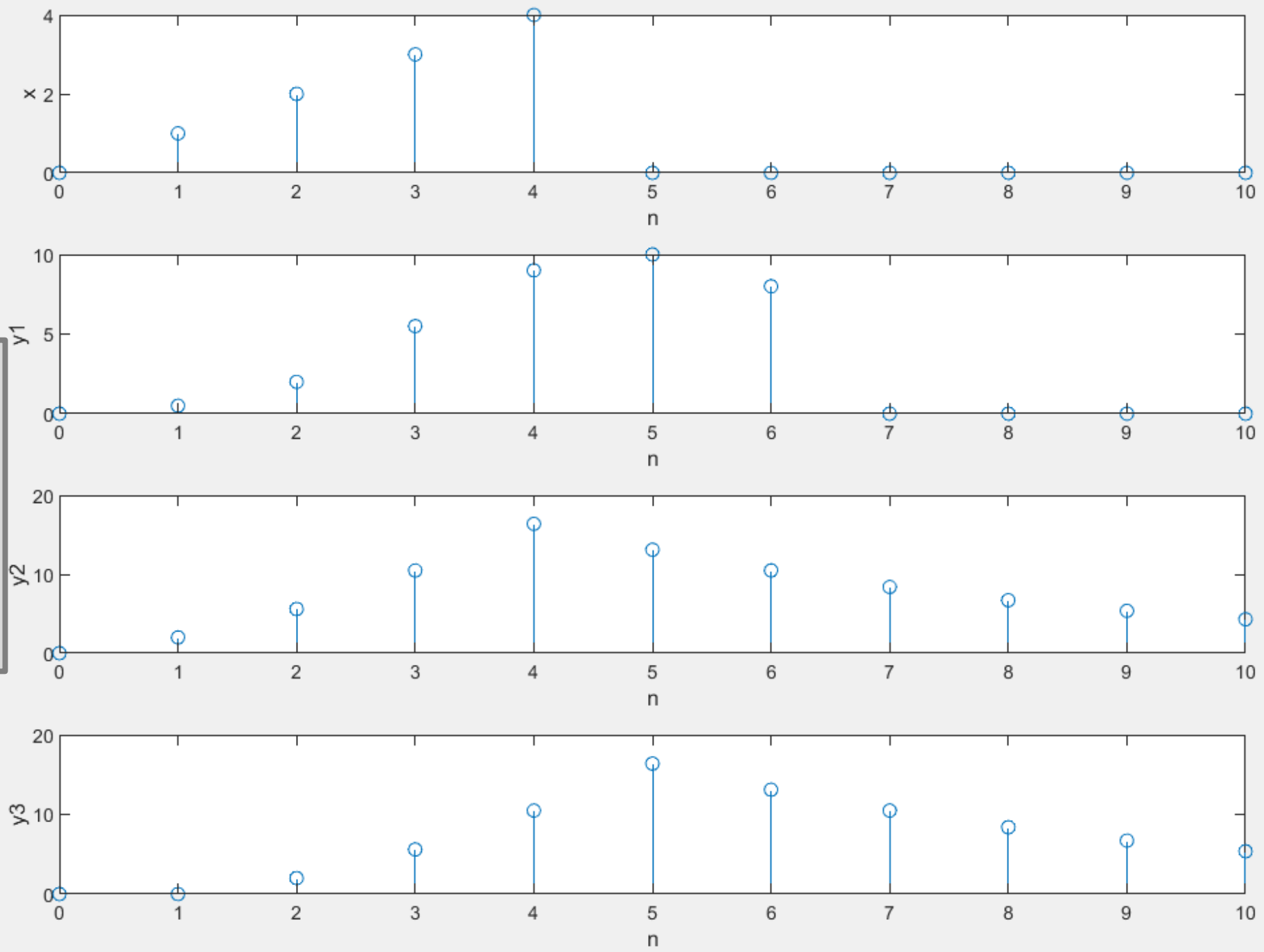
From Lab Book, 2.2)

```
n = 0:10;
x = n .* (n >= 1 & n <= 4);

y1 = filter(b1, a1, x);

y2 = filter(b2, a2, x);

y3 = filter(b3, a3, x);
```



Tutorial: filter / connect to conv

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The function `filter` can also be used to perform discrete-time convolution. Consider the class of systems satisfying Eq. (2.7) when $a_k = \delta[k]$. In this case, Eq. (2.7) becomes

$$y[n] = \sum_{m=0}^{M} b_m x[n-m]\,. \tag{2.9}$$

If we define the following finite-length signal

$$b[m] = \begin{cases} b_m\,, & 0 \le m \le M\,, \\ 0\,, & \text{otherwise}\,, \end{cases}$$

then Eq. (2.9) can be rewritten as

$$y[n] = \sum_{m=0}^{M} b[m]x[n-m] = \sum_{m=-\infty}^{\infty} b[m]x[n-m]\,. \tag{2.10}$$



Tutorial: filter / connect to conv

From Lab Book, 2.2) e, f, g

$$x[n] = \begin{cases} 1, & 0 \le n \le 5, \\ 0, & \text{otherwise}. \end{cases} \tag{2.5}$$

$$h[n] = \begin{cases} n, & 0 \le n \le 5, \\ 0, & \text{otherwise}. \end{cases} \tag{2.6}$$

```
nx = 0:5;
x = [1, 1, 1, 1, 1, 1];

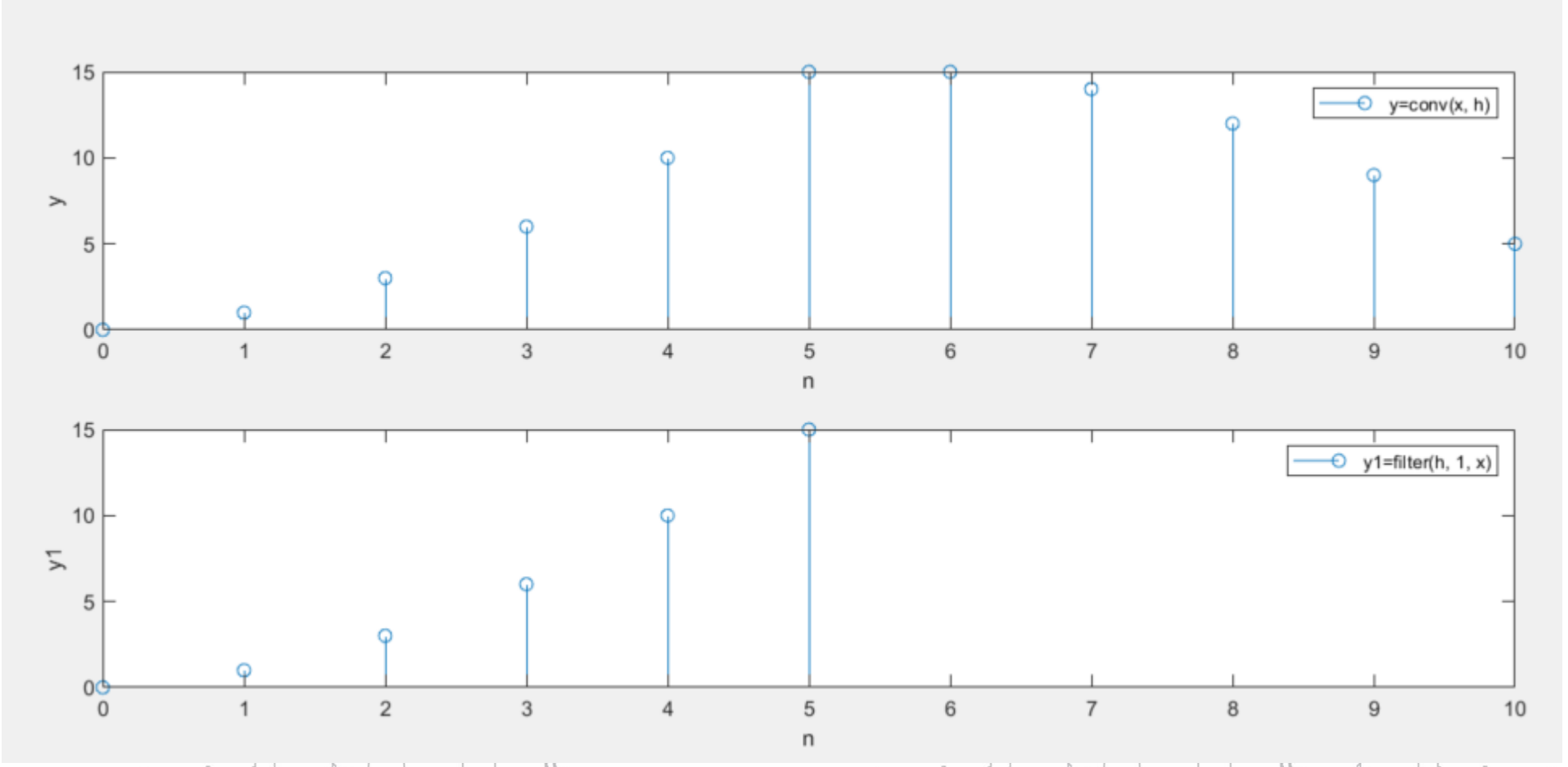
nh = 0:5;
h = nh;
```

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} h[k]\, x[n-k]$$

$$= x[n-1] + 2\,x[n-2] + 3\,x[n-3] + 4\,x[n-4] + 5\,x[n-5]$$

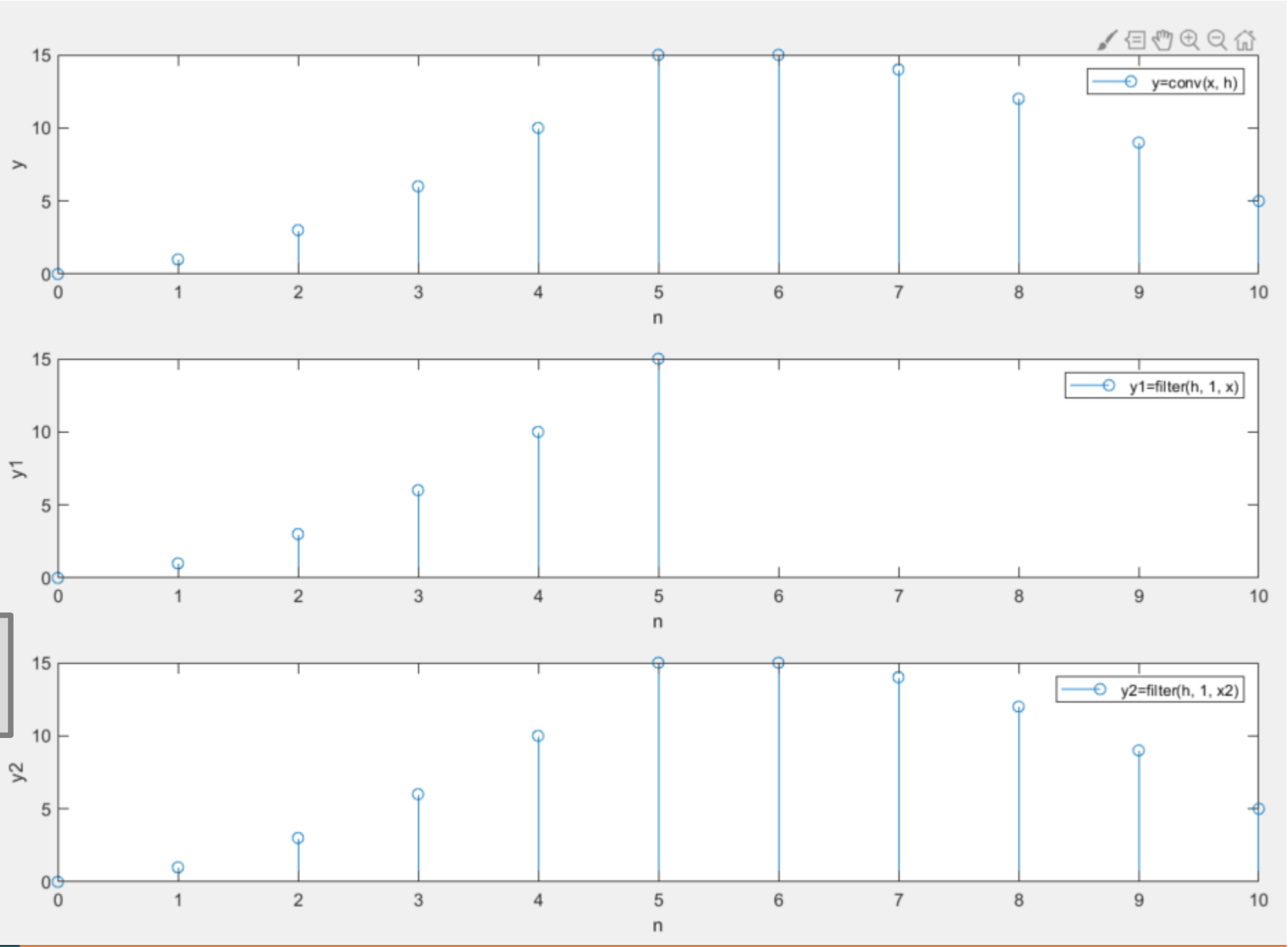
```
ny = (nx(1) + nh(1)) : (nx(end) + nh(end));
y = conv(x, h);
y1 = filter(h, 1, x);
```

Tutorial: filter / connect to conv



Tutorial: fil

```
x2 = [x, zeros(1, 5)];
y2 = filter(h, 1, x2);
```



Appendix: FIR, IIR

FIR (Finite-length Impulse Response) System
IIR (Infinite-length Impulse Response) System

$w = conv(u, v)$	$y = filter(b, a, x)$
Signal of finite duration	Signal of finite duration
FIR	FIR and IIR
nw $= nu(1)$ $+ nv(1):nu(end)$ $+ nv(end)$	$ny = nx$ Truncated output

DT LTI: Commutative, Associative, Distributive

From Lab Book, 2.4)

```
nx = [0:4];
x1 = ones(1, length(nx));

nh1 = [0:4];
h1 = [1, -1, 3, 0, 1];

nh2 = [0:4];
h2 = [0, 2, 5, 4, -1];
```

Basic Problems

(a). Many of the problems in this exercise will use the following three signals:

$$x_1[n] = \begin{cases} 1, & 0 \le n \le 4, \\ 0, & \text{otherwise}, \end{cases}$$

$$h_1[n] = \begin{cases} 1, & n = 0, \\ -1, & n = 1, \\ 3, & n = 2, \\ 1, & n = 4, \\ 0, & \text{otherwise}, \end{cases}$$

$$h_2[n] = \begin{cases} 2, & n = 1, \\ 5, & n = 2, \\ 4, & n = 3, \\ -1, & n = 4, \\ 0, & \text{otherwise}. \end{cases}$$

Define the MATLAB vector `x1` to represent $x_1[n]$ on the interval $0 \le n \le 9$, and the vectors `h1` and `h2` to represent $h_1[n]$ and $h_2[n]$ for $0 \le n \le 4$. Also define `nx1` and `nh1` to be appropriate index vectors for these signals. Make appropriately labeled plots of all the signals using `stem`.

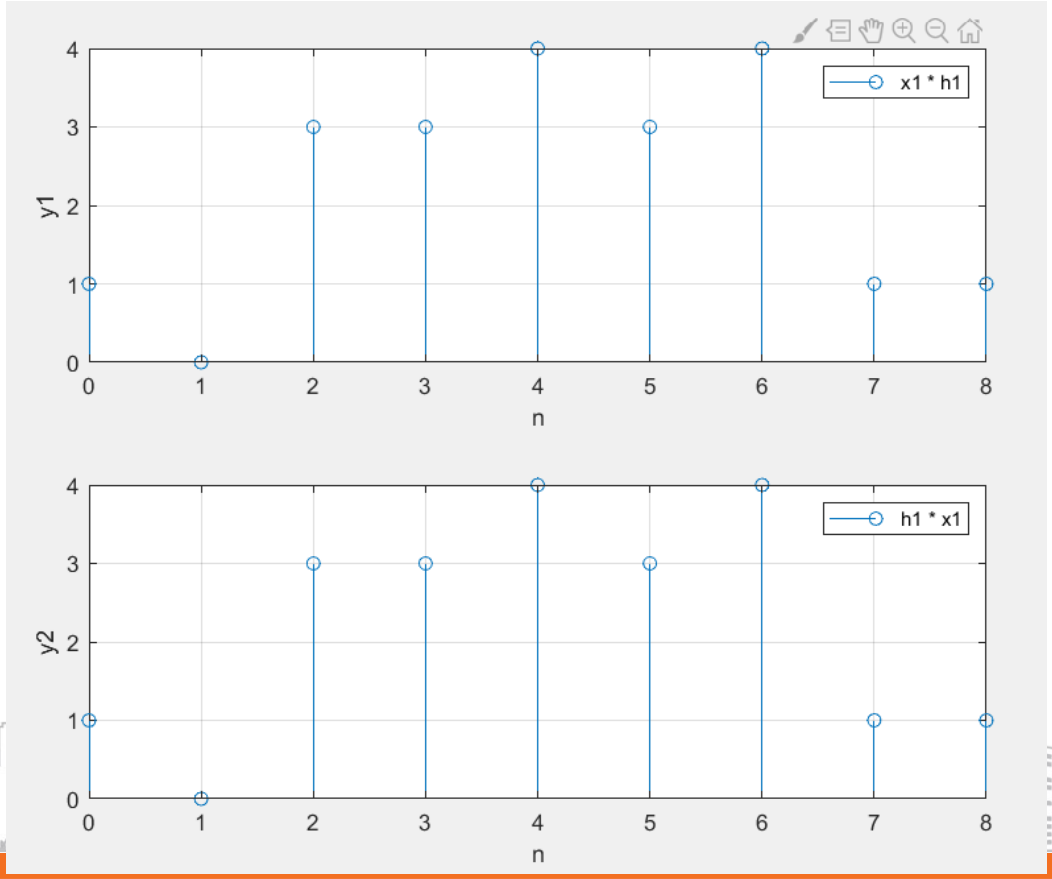
DT LTI: Commutative, Associative, Distributive

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From Lab Book, 2.4)

(b). The commutative property states that the result of a convolution is the same regardless of the order of the operands. This implies that the output of an LTI system with impulse response $h[n]$ and input $x[n]$ will be the same as the output of an LTI system with impulse response $x[n]$ and input $h[n]$. Use `conv` with `h1` and `x1` to verify this property. Is the output of `conv` the same regardless of the order of the input arguments?

```
ny = [(nx(1)+nh1(1)) :(nx(end)+nh1(end))];
y1 = conv(x1, h1);
y2 = conv(h1, x1);
```



DT LTI: Commutative, Associative, Distributive

From Lab Book, 2.4)

(c). Convolution is also distributive. This means that

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n] .$$

This implies that the output of two LTI systems connected in parallel is the same as one system whose impulse response is the sum of the impulse responses of the parallel systems. Figure 2.8 illustrates this property.

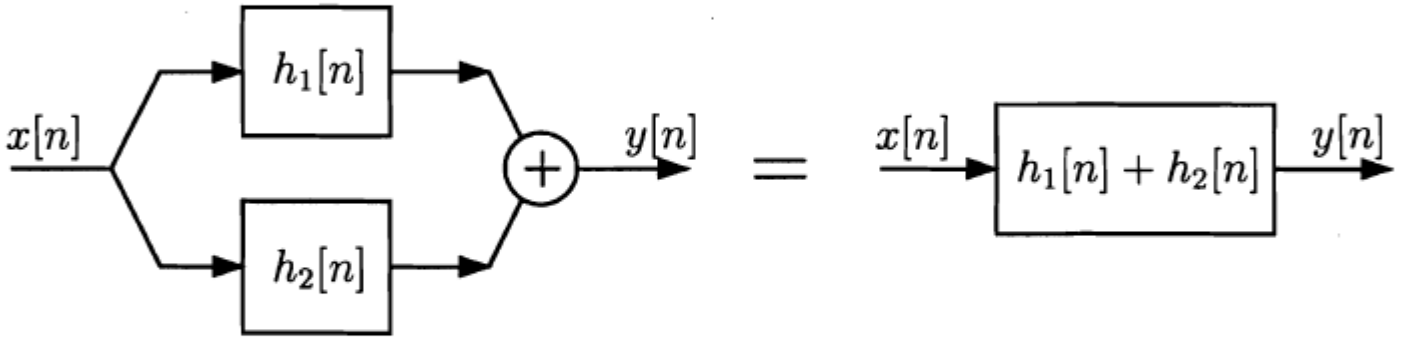


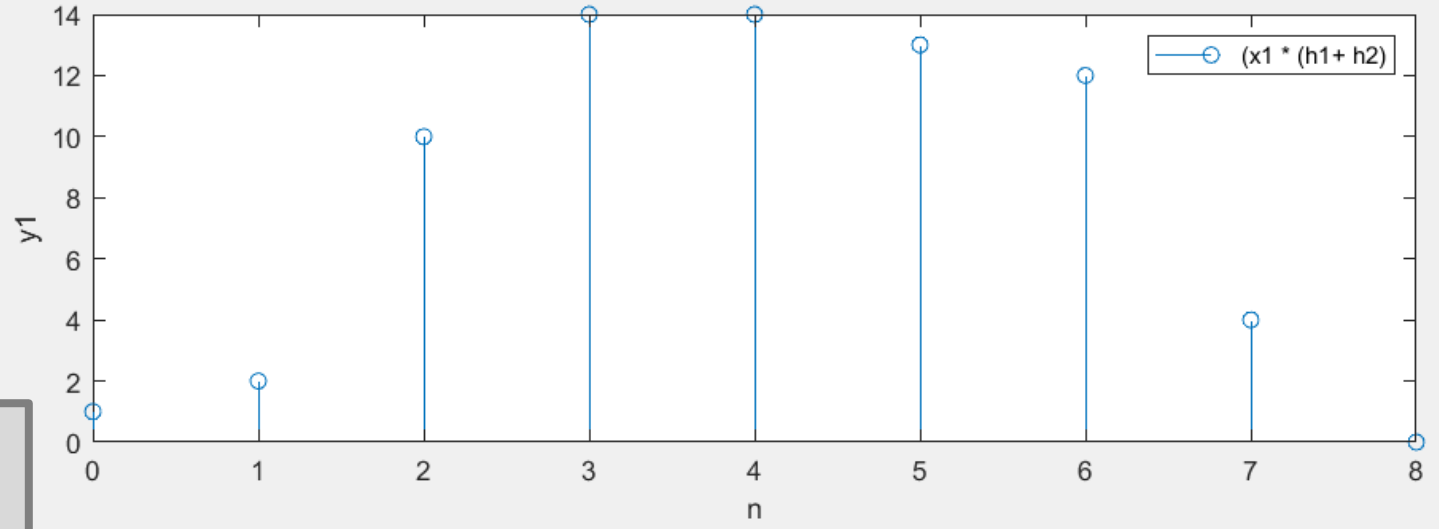
Figure 2.8. Distributive property of convolution.

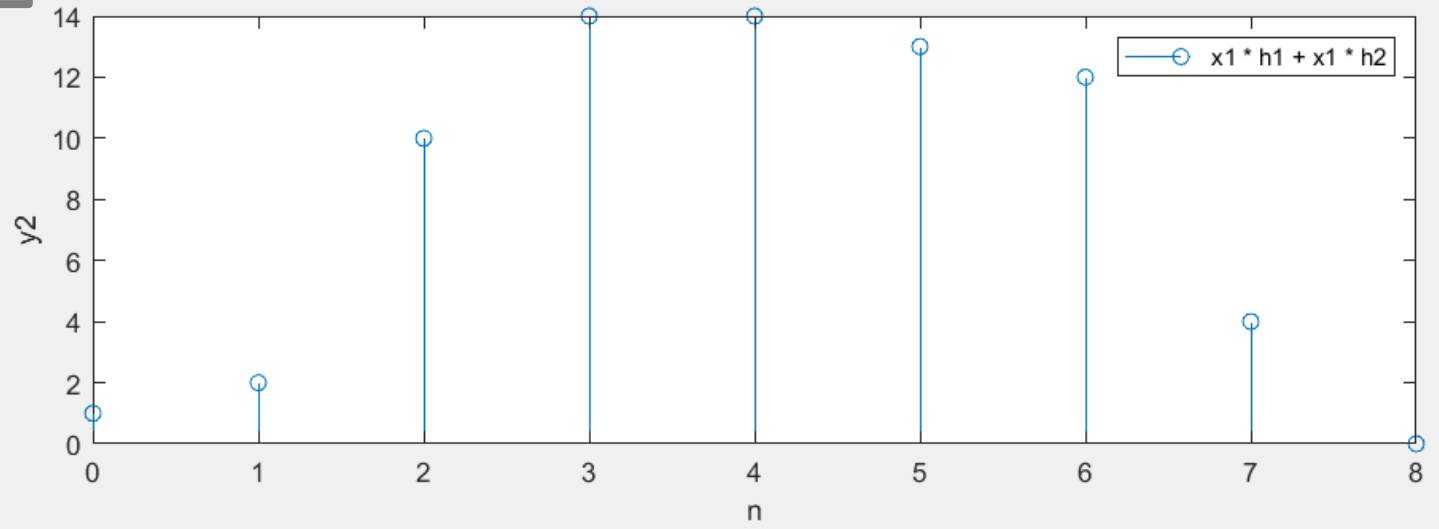
Verify the distributive property using `x1`, `h1` and `h2`. Compute the sum of the outputs of LTI systems with impulse responses $h_1[n]$ and $h_2[n]$ when $x_1[n]$ is the input. Compare this with the output of the LTI system whose impulse response is $h_1[n] + h_2[n]$ when the input is $x_1[n]$. Do these two methods of computing the output give the same result?

DT LTI: Commutative, Associative, Distributive

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```
ny = [(nx(1)+nh1(1)) :(nx(end)+nh1(end))];
y1 = conv(x1, (h1 + h2));
y2 = conv(x1, h1) + conv(x1, h2);
```





DT LTI: Commutative, Associative, Distributive

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From Lab Book, 2.4)

(d). Convolution also possesses the associative property, i.e.,

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n]) .$$

This property implies that the result of processing a signal with a series of LTI systems is equivalent to processing the signal with a single LTI system whose impulse response is the convolution of all the individual impulse responses of the connected systems. Figure 2.9 illustrates this property for the case of two LTI systems connected in series.

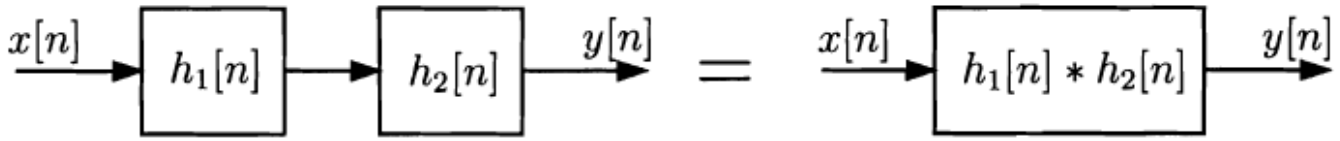


Figure 2.9. Associative property of convolution.

Use the following steps to verify the associative property using `x1`, `h1` and `h2`:

- Let $w[n]$ be the output of the LTI system with impulse response $h_1[n]$ shown in Figure 2.9. Compute $w[n]$ by convolving $x_1[n]$ and $h_1[n]$.
- Compute the output $y_{d1}[n]$ of the whole system by convolving $w[n]$ with $h_2[n]$.
- Find the impulse response $h_{\text{series}}[n] = h_1[n] * h_2[n]$.
- Convolve $x_1[n]$ with $h_{\text{series}}[n]$ to get the output $y_{d2}[n]$.

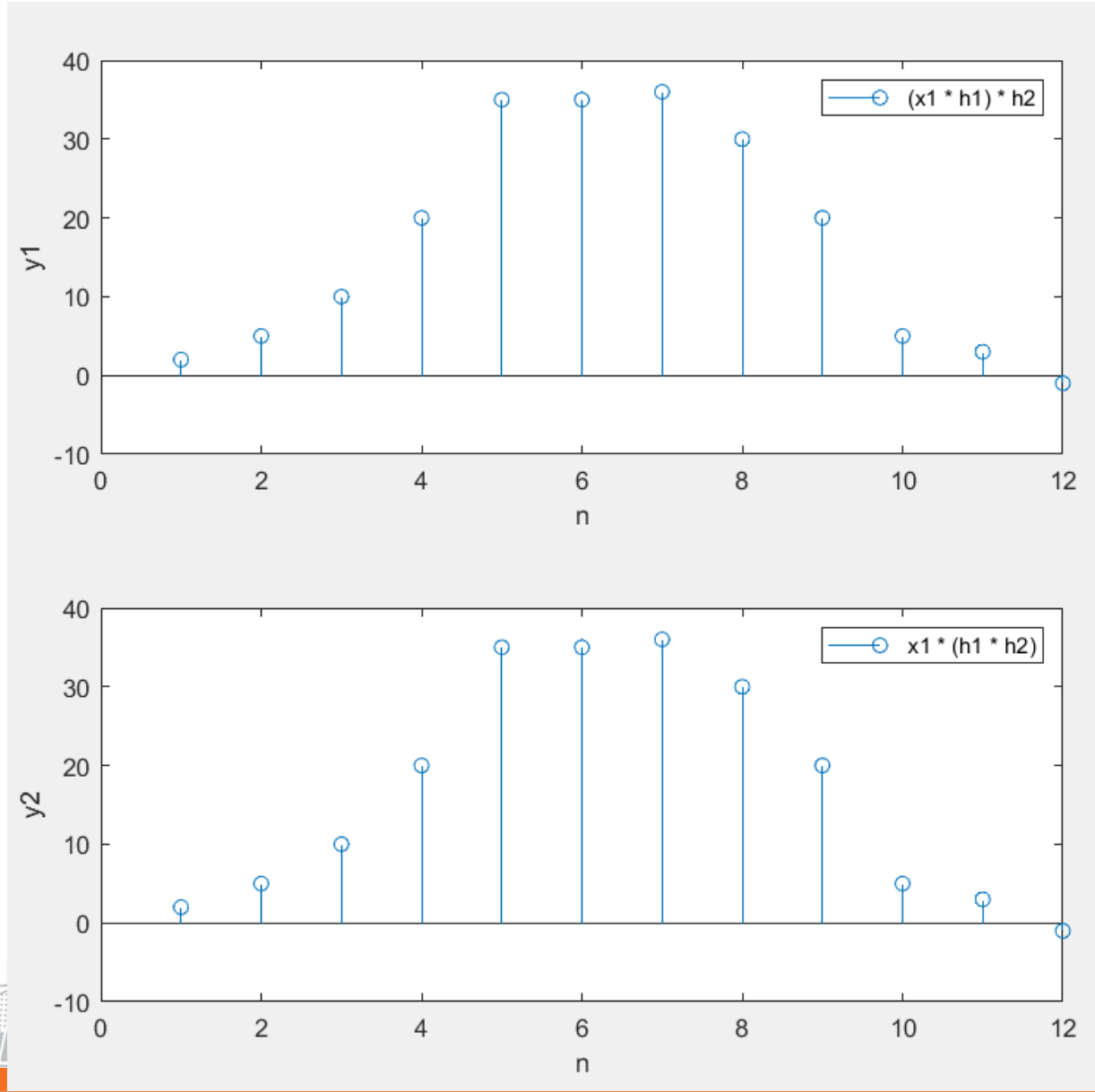
Compare $y_{d1}[n]$ and $y_{d2}[n]$. Did you get the same results when you process $x_1[n]$ with the individual impulse responses as when you process it with $h_{\text{series}}[n]$?

DT LTI: Commutative, Associative, Distributive

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```
ny = [(nx(1) + nh1(1) + nh2(1)) :
(nx(end) + nh1(end) + nh2(end))];

y1 = conv(conv(x1, h1), h2);
y2 = conv(x1, conv(h1, h2));
```



DT LTI: linearity and Time-Invariance

From Lab Book, 2.5)

w[n] = conv(x, h), **showcase**

课堂作业

```
nh = [0, 1, 2];
h = [1, -1, -1];
```

```
nx = [0:5];
x1 = [1, zeros(1, length(nx)-1)];
x2 = [0, 1, zeros(1, length(nx)-2)];
x3 = x1 + 2 .* x2;
```

Basic Problems

Consider the following three systems:

$$\begin{aligned} &\text{System 1:} && w[n] = x[n] - x[n-1] - x[n-2]\,, \\ &\text{System 2:} && y[n] = \cos(x[n])\,, \\ &\text{System 3:} && z[n] = n\,x[n]\,, \end{aligned}$$

where $x[n]$ is the input to each system, and $w[n]$, $y[n]$, and $z[n]$ are the corresponding outputs.

(a). Consider the three inputs signals $x_1[n] = \delta[n]$, $x_2[n] = \delta[n-1]$, and $x_3[n] = (\delta[n] + 2\,\delta[n-1])$. For System 1, store in `w1`, `w2`, and `w3` the responses to the three inputs. The vectors `w1`, `w2`, and `w3` need to contain the values of $w[n]$ only on the interval $0 \le n \le 5$. Use `subplot` and `stem` to plot the four functions represented by `w1`, `w2`, `w3`, and `w1+2*w2` within a single figure. Make analogous plots for Systems 2 and 3.

Q: Linearity?

(b). State whether or not each system is linear. If it is linear, justify your answer. If it is not linear, use the signals plotted in Part (a) to supply a counter-example.

Q: Time-Invariance?

(c). State whether or not each system is time-invariant. If it is time-invariant, justify your answer. If it is not time-invariant, use the signals plotted in Part (a) to supply a counter-example.

DT LTI: linearity and Time-Invariance

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```
nw = [(nx(1) + nh(1)) :
(nx(end) + nh(end))];

w1 = conv(x1, h);

w2 = conv(x2, h);

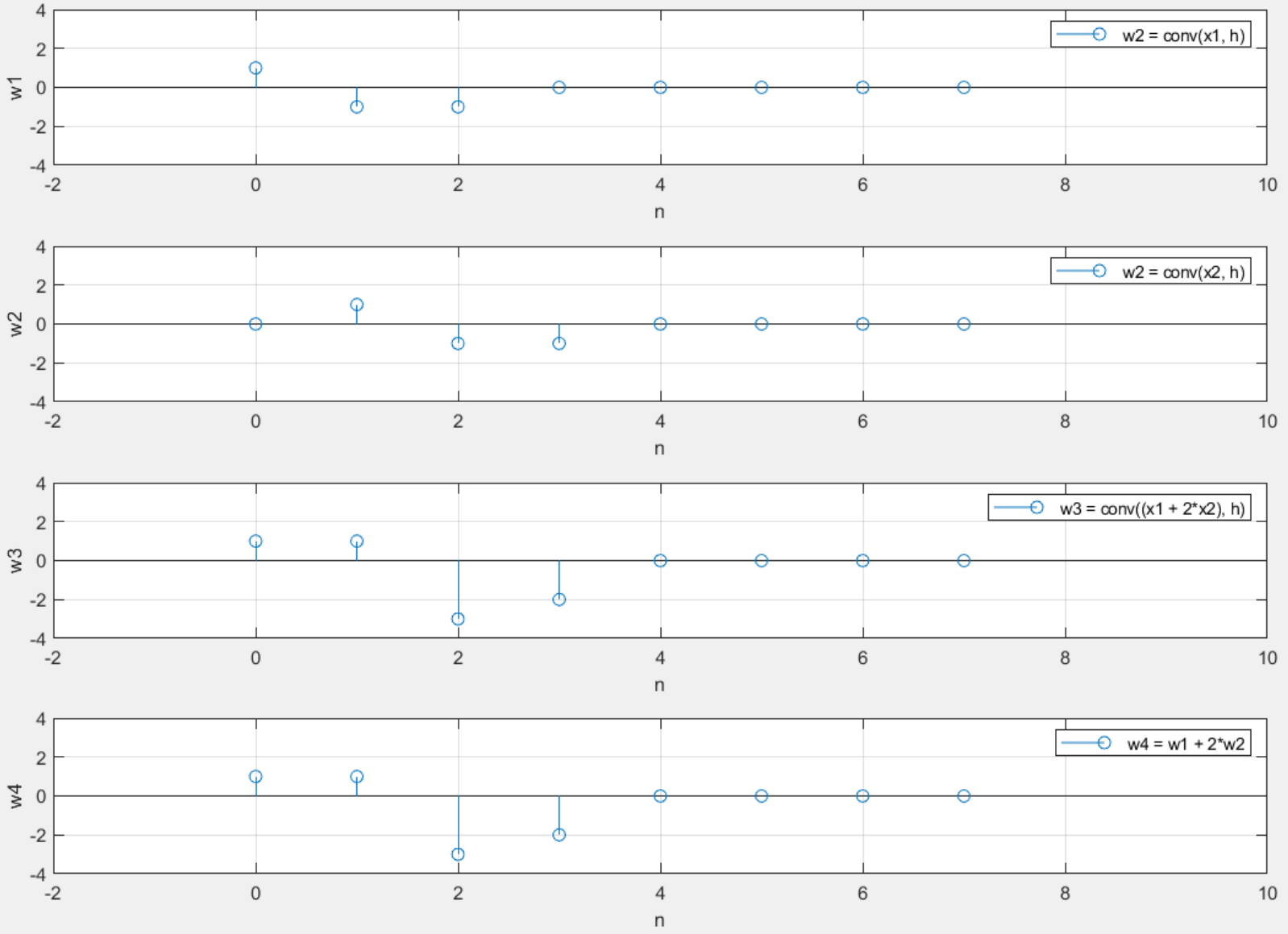
w3 = conv(x3, h);

w4 = w1 + 2 .* w2;
```

For System 1

w1 vs w2
=> Time-Invariance: YES

w3 vs w4
=> Linearity: YES



DT LTI: linearity and Time-Invariance

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From Lab Book, 2.5)

```
function y = lab2_5defg_sys2(nx, x, y0)
%  y[n] = (3/5)^n * y[n - 1] + x[n]
y = zeros(1, length(x));
y(1) = x(1) + (3/5)^nx(1) * y0;
for i = 2:length(y)
    y(i) = x(i) + (3/5)^nx(i) * y(i-1);
end
```

```
nx = [0:19];
d = [1, zeros(1, length(nx)-1)];
u = ones(1, length(nx));

%  nh2 = nx
h2 = lab2_5defg_sys2(nx, d, 0);

%  ns2 = nx
s2 = lab2_5defg_sys2(nx, u, 0);
```

Intermediate Problems

In these problems, you will be asked to consider how the impulse response can be used to calculate the step response of an LTI system. Consider the two causal systems defined by the following linear difference equations:

课堂作业 Showcase

System 1: $y_1[n] = (3/5)\, y_1[n-1] + x[n]\,,$

System 2: $y_2[n] = (3/5)^n\, y_2[n-1] + x[n]\,,$

where each system satisfies initial rest conditions. Initial rest conditions state that if $x[n] = 0$ for $n \leq n_0$, then $y[n] = 0$ for $n \leq n_0$. Define $h_1[n]$ and $h_2[n]$ to be the responses of Systems 1 and 2, respectively, to the signal $\delta[n]$.

(d). Calculate $h_1[n]$ and $h_2[n]$ on the interval $0 \leq n \leq 19$, and store these responses in `h1` and `h2`. Plot each response using `stem`. Hint: The `filter` function can be used to calculate `h1`. However, System 2 is described by a difference equation with non-constant coefficients; therefore, you must either determine `h2` analytically or use a `for` loop rather than `filter` to calculate `h2`.

(e). For each system, calculate the unit step response on the interval $0 \leq n \leq 19$, and store the responses in `s1` and `s2`. Again, `filter` can be used only to calculate the step response of System 1. Use a `for` loop to calculate `s2`.

DT LTI: linearity and Time-Invariance

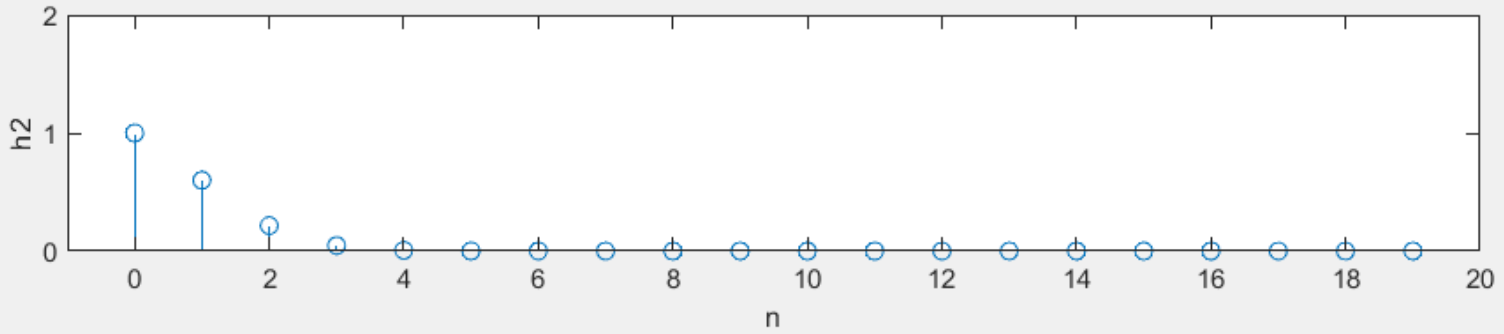
From Lab Book, 2.5)

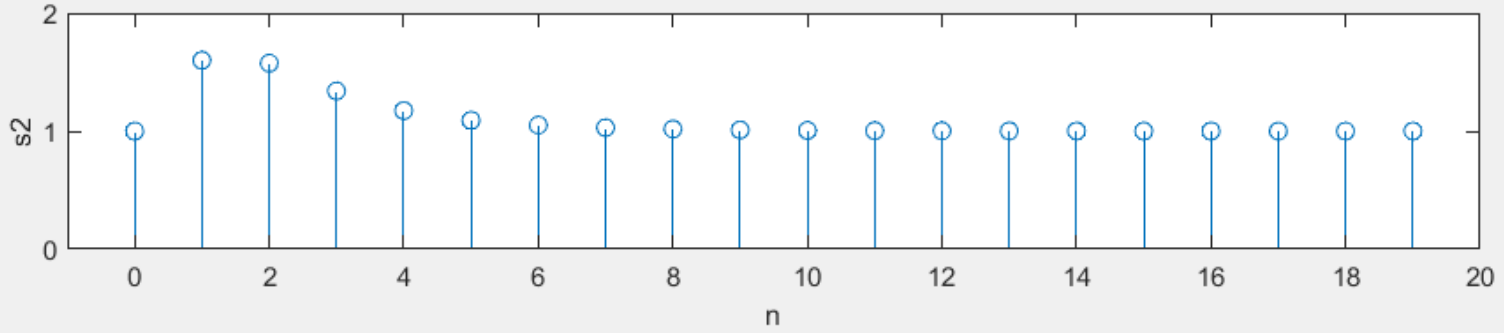
(f). Note that $h_1[n]$ and $h_2[n]$ are zero for $n \geq 20$ for all practical purposes. Thus `h1` and `h2` contain all we need to know about the response of each system to the unit impulse. Define $z_1[n] = h_1[n] * u[n]$ and $z_2[n] = h_2[n] * u[n]$, where $u[n]$ is the unit step function. Use `conv` to calculate $z_1[n]$ and $z_2[n]$ on the interval $0 \leq n \leq 19$, and store these calculations in the vectors `z1` and `z2`. You must first define a vector containing $u[n]$ over an appropriate interval, and then select the subset of the samples produced by `conv(h1,u)` and `conv(h2,u)` that represent the interval $0 \leq n \leq 19$. Since you have truncated two infinite-length signals, only a portion of the outputs of `conv` will contain valid sequence values. This issue was also discussed in Exercise 2.7 Part (c).

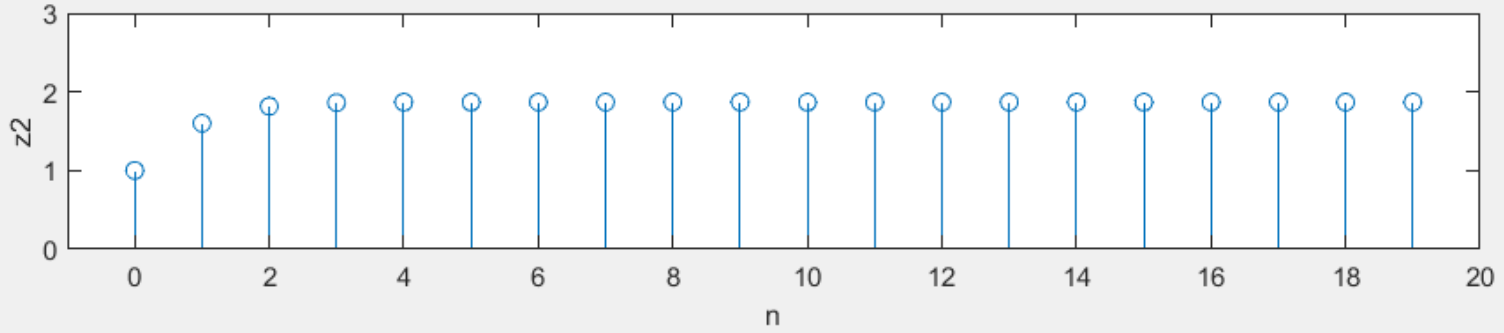
(g). Plot `s1` and `z1` on the same set of axes. If the two signals are identical, explain why you could have anticipated this similarity. Otherwise, explain any differences between the two signals. On a different set of axes, plot `s2` and `z2`. Again, explain how you might have anticipated any differences or similarities between these two signals.

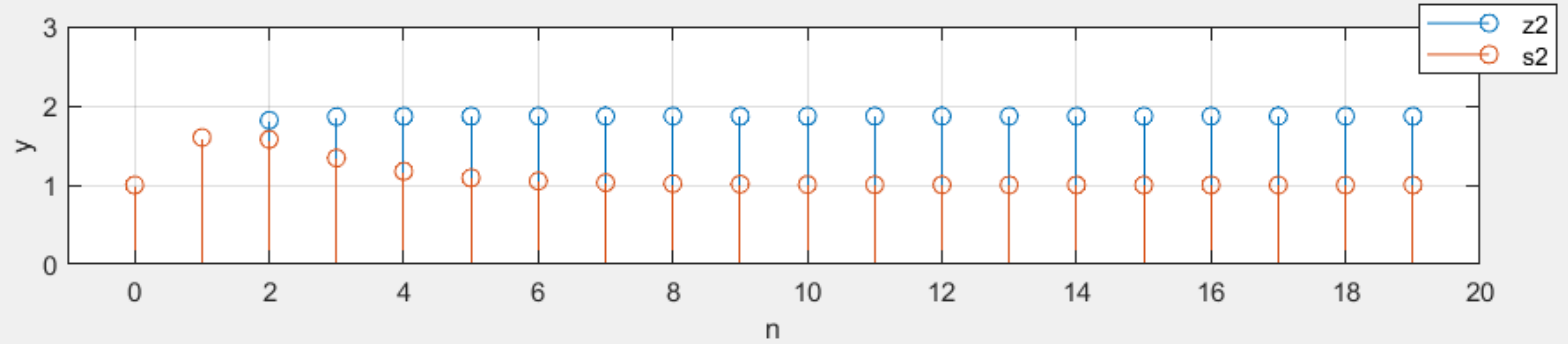
```
nz2 =  [(2*nx(1)) : (2*nx(end))];
z2 = conv(h2, u);
```

DT LTI: linearity ar









s2, z2 is different

=> System 2 is Non-Linearity

LTI System / 课堂练习1

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2.1: d), e)

2.2: g), h), i), j)

2.4: e), f)

2.5:

- a) ~ c) for system 2
- d) ~ g) for system 1

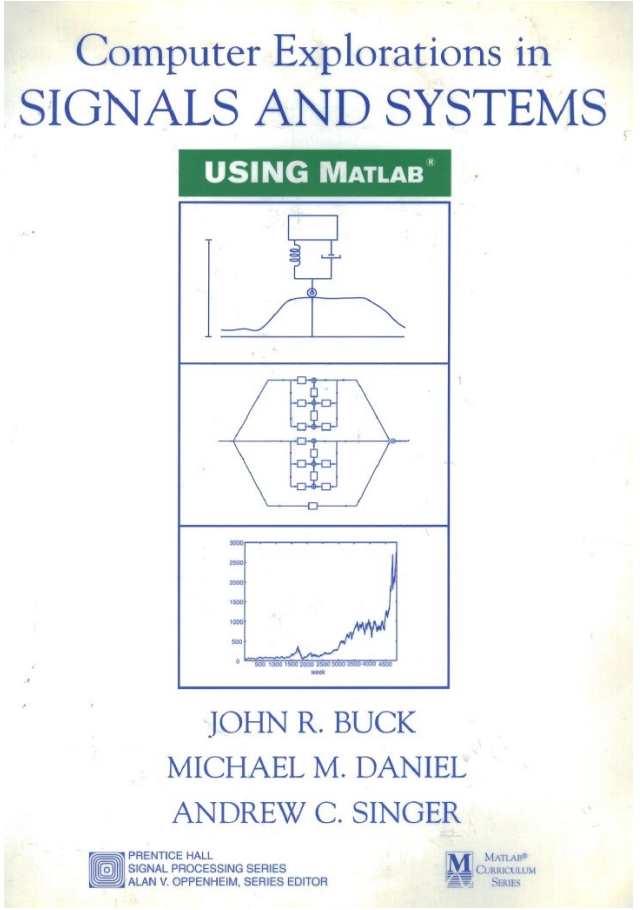
2.10:

- a) ~ e), required
- f), optional, as bonus

绿色部分：不用提交，建议练习

红色部分：本次课堂作业

其他：下次课堂作业



样本 lineup.mat 下载链接:
https://pan.baidu.com/s/1M3QxJ-gCiVTteWqStO25gA?pwd=r9gp
提取码: r9gp

Outline

- Discrete-Time LTI Systems
- **Continuous-Time LTI Systems**
- Showcase: Echo Cancellation

2 Linear Time-Invariant Systems **19**

2.1 Tutorial: `conv` . . . 20
2.2 Tutorial: `filter` . . . 22
2.3 Tutorial: `lsim` with Differential Equations . . . 26
2.4 Properties of Discrete-Time LTI Systems . . . 29
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2.6 Noncausal Finite Impulse Response Filters . . . 34
2.7 Discrete-Time Convolution . . . 36
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2.10 Echo Cancellation via Inverse Filtering . . . 44

CT: Convolution Integral

$$\delta_\Delta(t-k\Delta) \rightarrow \boxed{S} \rightarrow \hat{h}_{k\Delta}(t) \quad (k=\cdots,-1,0,1,\cdots)$$

S is linear

$$\hat{x}(t)=\sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t-k\Delta)\Delta \rightarrow \boxed{S} \rightarrow \hat{y}(t)=\sum_{k=-\infty}^{+\infty} x(k\Delta)\,\hat{h}_{k\Delta}(t)\Delta$$

$$x(t)=\lim_{\Delta\to 0}\hat{x}(t)=\int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau \qquad y(t)=\lim_{\Delta\to 0}\hat{y}(t)=\int_{-\infty}^{+\infty} x(\tau)h_\tau(t)d\tau$$

S is time-invariant

$$h_\tau(t)=h(t-\tau)$$

$$x(t)=\int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)d\tau \rightarrow \boxed{S} \rightarrow y(t)=\int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau$$

CT: Convolution Integral

From Lab Book, 2.8)

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)\, h\,(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{+\infty} e^{-2\tau} u(\tau)\, u(t-\tau) d\tau$$

$$= \int_{0}^{t} e^{-2\tau} d\tau = \frac{1}{2} - \frac{1}{2} e^{-2t}$$

Note that $y_\Delta(t)$ is not necessarily itself piecewise constant, but rather the convolution of two piecewise constant signals.

For this exercise, you will approximate the convolution of $x(t) = e^{-2t}u(t)$ with $h(t) = u(t)$ using piecewise constant functions for several different values of Δ. As $\Delta \to 0$, this approximation will converge to the analytic convolution of these two functions.

Basic Problem

(a). Find $y(t) = x(t) * h(t)$ analytically for the functions $x(t)$ and $h(t)$ defined above. You will need to know the result of this convolution so you can compare it to the numerical approximations you will be computing in this exercise.

(b). Starting from the convolution integral, Eq. (2.17), derive that in general

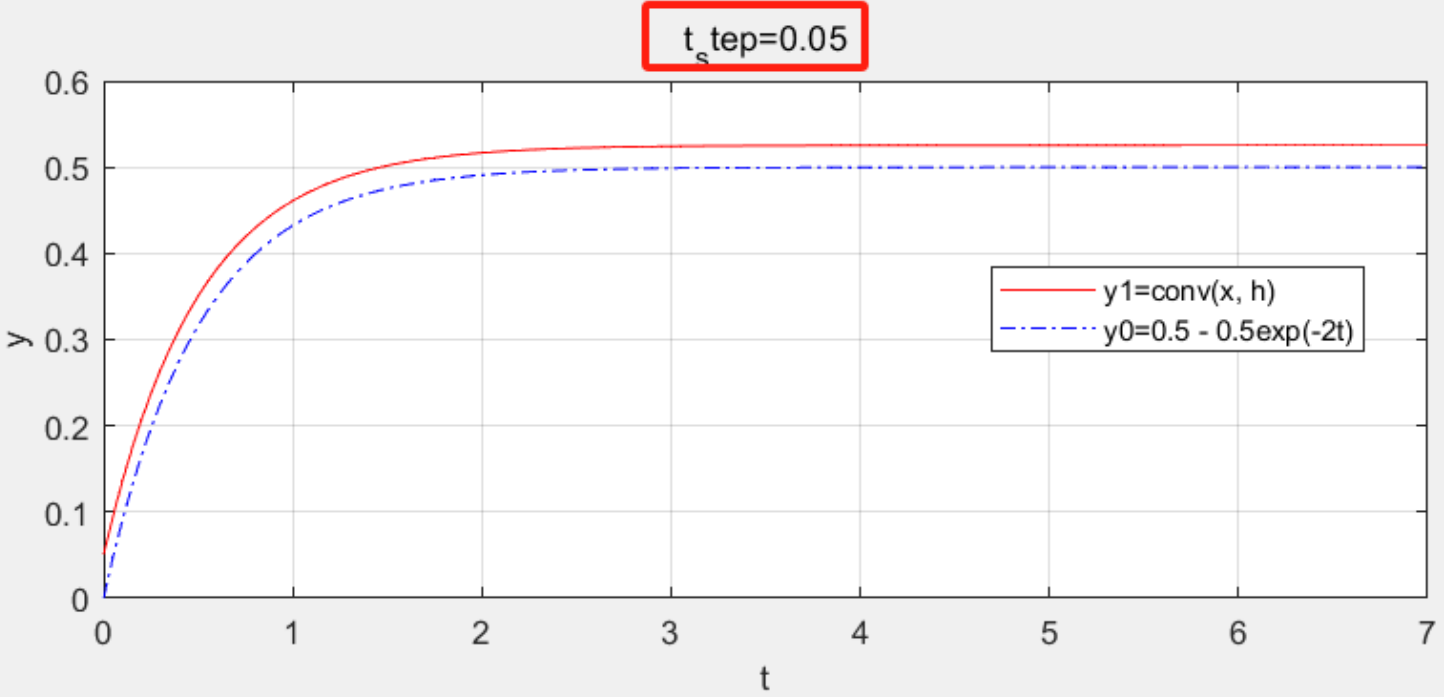
$$y_\Delta(n\Delta) = \Delta \sum_{k=-\infty}^{\infty} x(k\Delta)\, h\big((n-k)\Delta\big). \qquad (2.18)$$

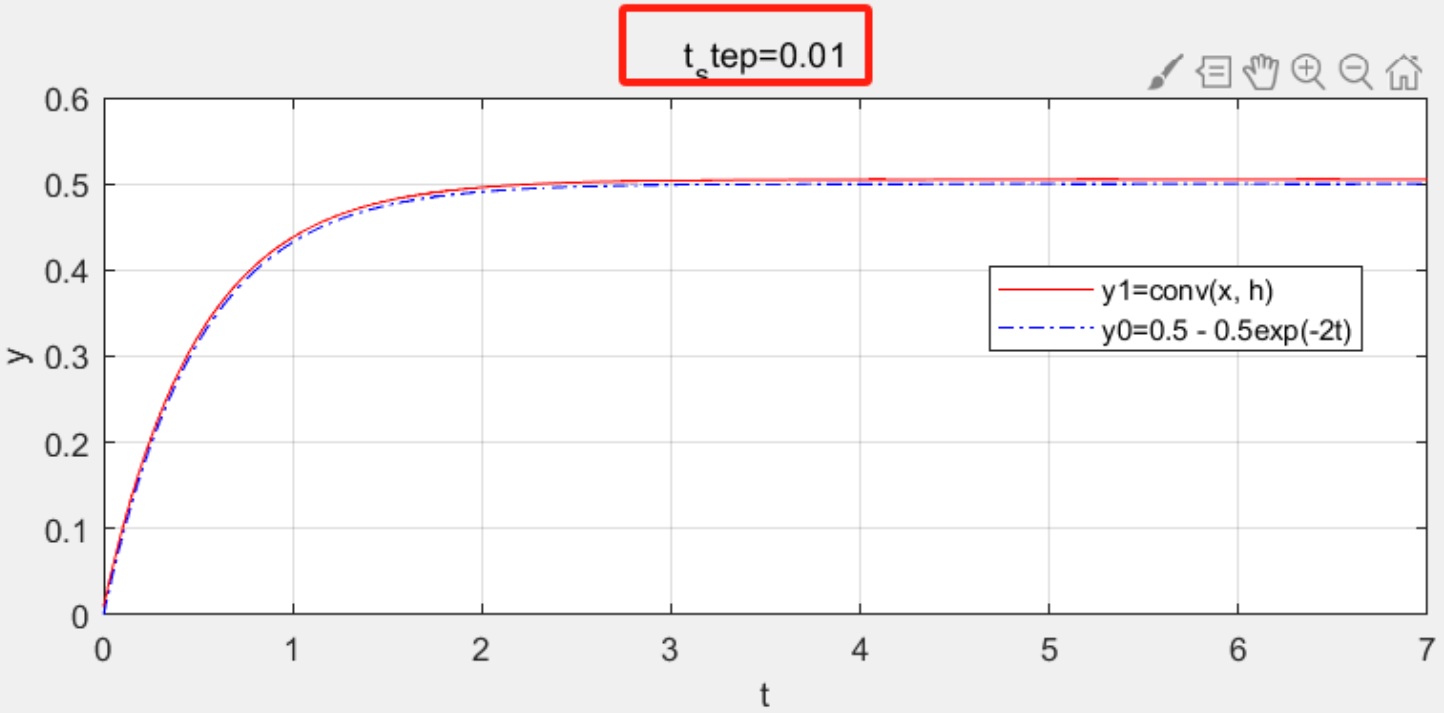
The expression you derived in Part (b) can be used to approximate $y(t)$. If the signals $x_\Delta(t), h_\Delta(t)$, and $y_\Delta(t)$ are represented by MATLAB vectors containing the values of the signals at times $t = n\Delta$, then Eq. (2.18) is a discrete convolution sum of the MATLAB vectors. By computing this discrete convolution as the value of Δ decreases, you can obtain closer approximations to $y(t)$.

CT: Convolution Inte

```
t_step = 0.01;
t = [0:t_step:7];
x = exp(-2 * t) .* (t >= 0);
h = 1.0 .* (t >= 0);

y1 = conv(x, h) * t_step;
y0 = 0.5 - 0.5 * exp(-2 * t);
```





CT LTI: Pulse Response

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Analytically derive

When X(t) is unit step

when t >= 0

$$y(t) = Ce^{-3t} + \frac{1}{3}$$

when t = 0

$$y(t) = 0 \Rightarrow C = -\frac{1}{3}$$

Then

$$s(t) = -\frac{1}{3}e^{-3t} + \frac{1}{3}$$

X(t) is impulse

$$h(t) = e^{-3t}$$

For this exercise, you will consider the causal LTI system whose input $x(t)$ and output $y(t)$ satisfy the following first-order linear differential equation:

$$\frac{dy(t)}{dt} + 3\,y(t) = x(t)\,. \tag{2.20}$$

You will consider the response of this system to pulses of various shapes and durations, and will discover how accurately $h^{\Delta}(t)$ characterizes the system for various values of Δ.

Basic Problems

(a) Analytically derive the unit step response $s(t)$ and the unit impulse response $h(t)$ of the causal LTI system defined by Eq. (2.20). Hint: The impulse response is equal to $ds(t)/dt$. Store in `h` and `s` the values of $h(t)$ and $s(t)$ at each time sample in the vector `t=[-1:0.05:4]`. Plot both `h` and `s` versus `t`.

(b) Use `step` and `impulse` to verify the functions that you computed in Part (a). The functions `step(b,a,t)` and `impulse(b,a,t)` are explained in Tutorial 2.3, the tutorial for `lsim`. The vectors `b` and `a` are determined by the coefficients in Eq. (2.20). For the time vector `t`, use `t=[0:0.05:4]`.

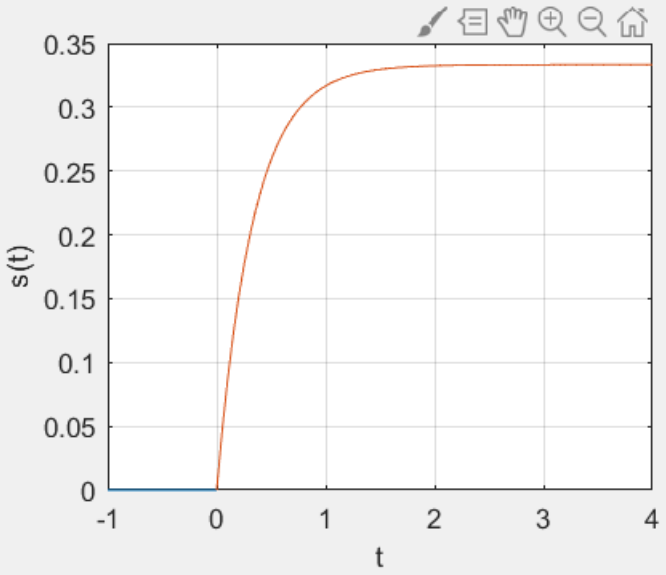
CT LTI: Pulse Response

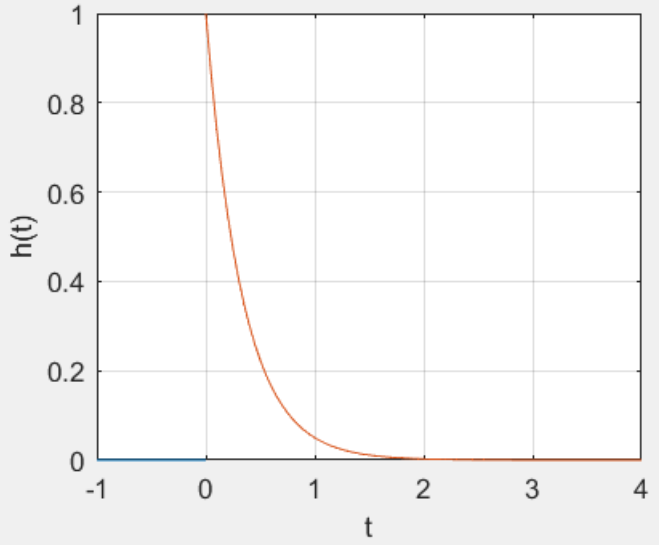
Matlab verify: step(...), impulse(...)

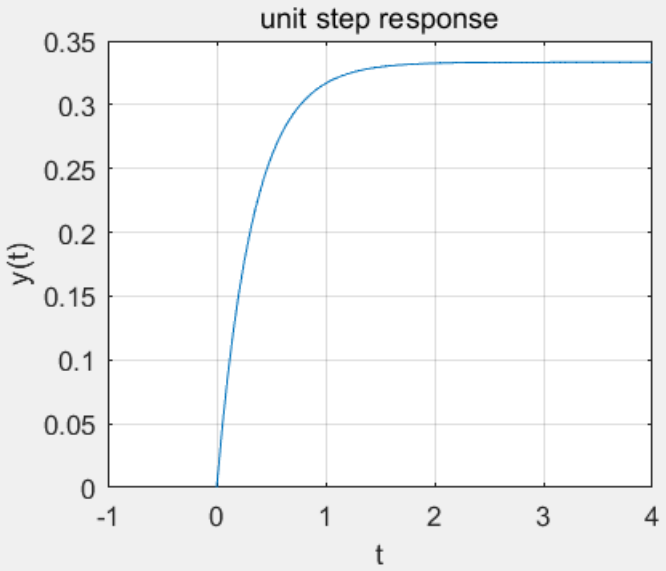
```
t_step = 0.05;
t0 = [-1:t_step:0];
t = [0:t_step:4];

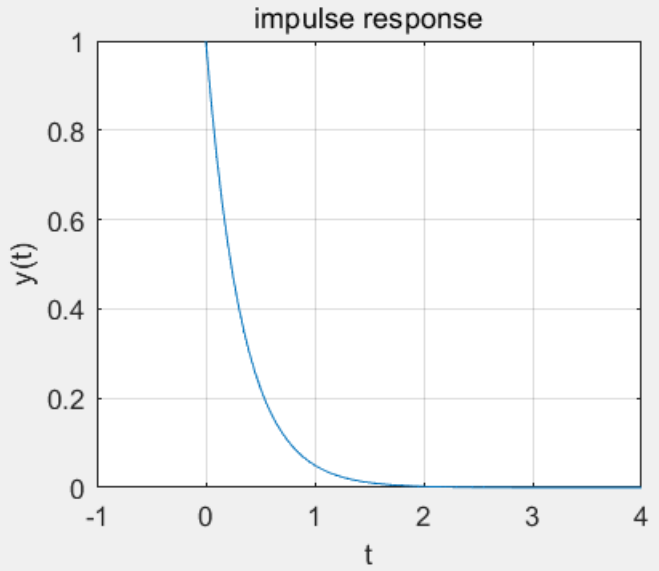
y0 = zeros(1, length(t0));
s = -1/3 * exp(-3 .* t) + 1 / 3;
h = exp(-3 .* t);

y_step = step([1], [1, 3], t);
y_impulse = impulse([1], [1, 3], t);
```









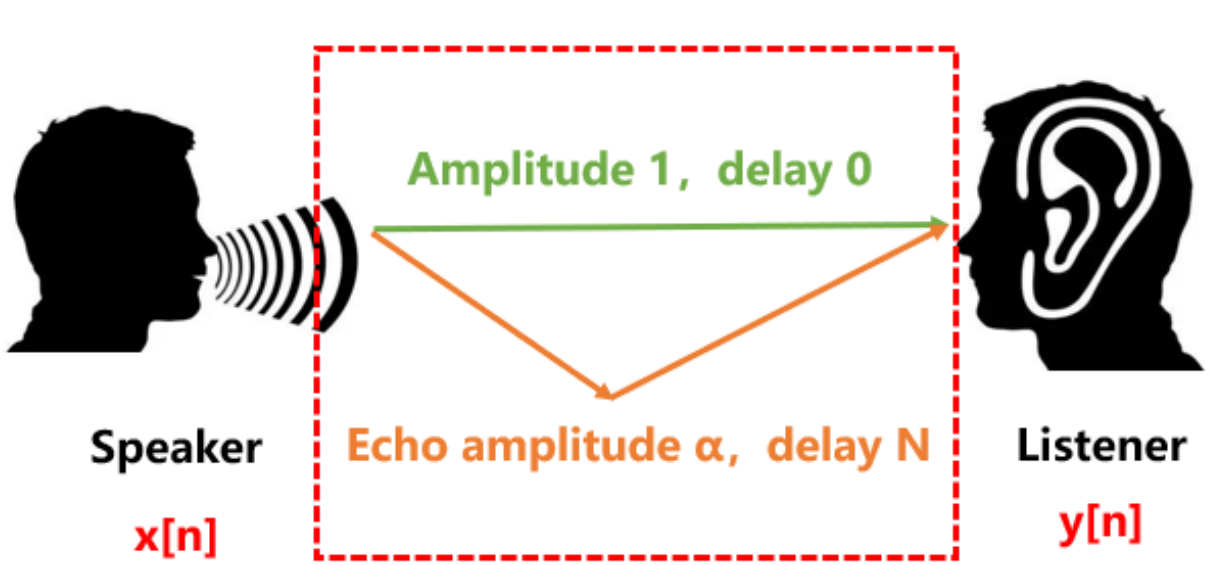
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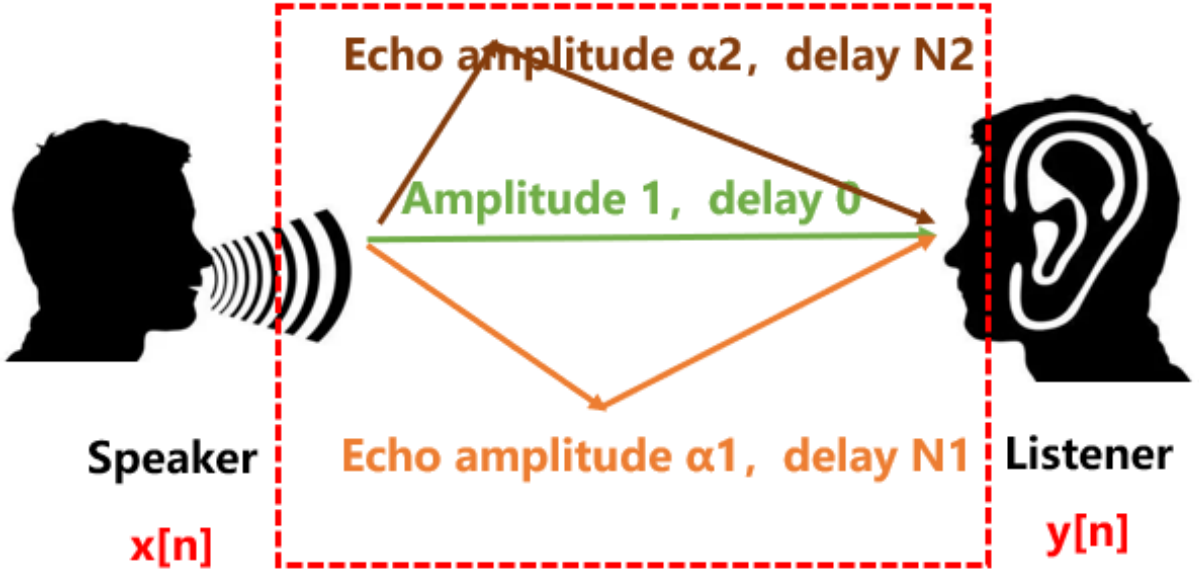
Echo Cancellation: Background

Basic Problem: Removing an echo from a recording of a speech signal
=> Based on known/unknown channel model, and given y[n], how to get back x[n] ?



Voice Channel, as LTI system

$y[n] = x[n] + \alpha\, x[n-N]$



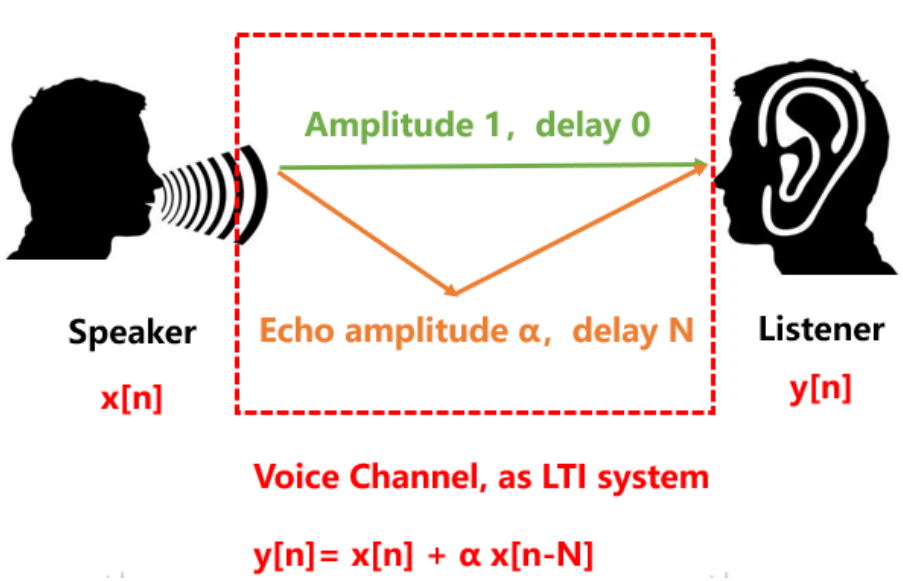
Voice Channel, as LTI system

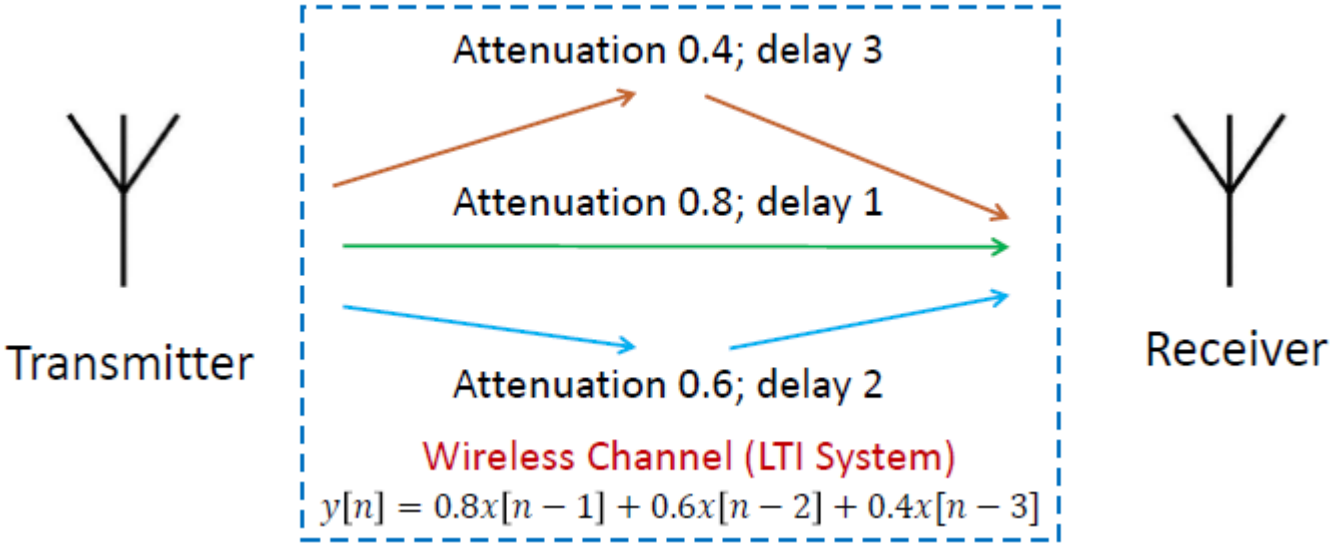
$y[n] = x[n] + \alpha 1\, x[n-N1] + \alpha 2\, x[n-N2]$

Echo Cancellation: Background

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Similar as: Simplified Wireless Channel

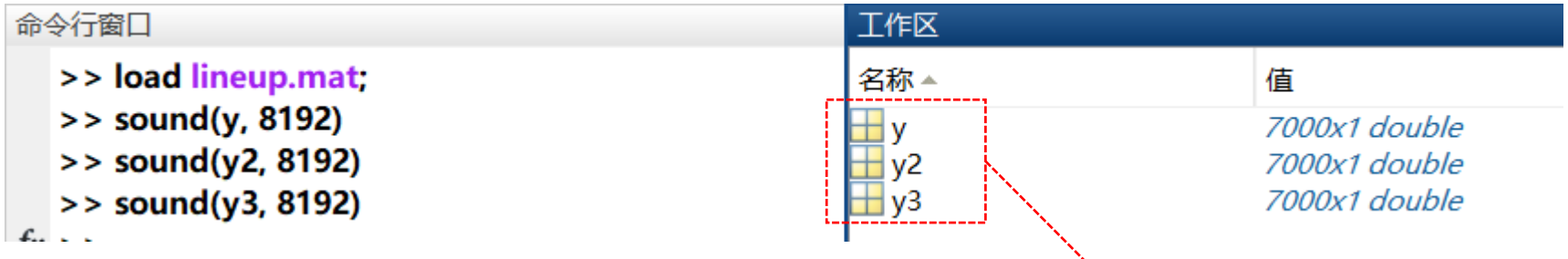




Echo Cancellation: Background

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How to get Y[n] ?



Problem #1: …

Problem #2: …

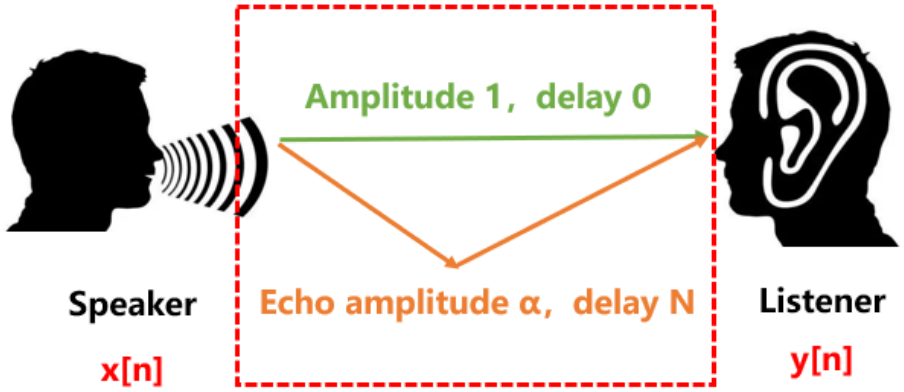
Problem #3: …

样本 lineup.mat 下载链接:
https://pan.baidu.com/s/1M3QxJ-gCiVTteWqStO25gA?pwd=r9gp
提取码: r9gp

Echo Cancellation: Problem #1

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Problem #1, when only has 1 echo, and α, N given, to get back x[n]?



Voice Channel, as LTI system

y[n]= x[n] + α x[n-N]

You should hear the phrase "line up" with an echo. The signal $y[n]$, represented by the vector y, is of the form

$$y[n] = x[n] + \alpha x[n - N], \tag{2.21}$$

For the problems in this exercise, you will use the value of the echo time, $N = 1000$, and the echo amplitude, $\alpha = 0.5$.

Basic Problems

(a). In this exercise you will remove the echo by linear filtering. Since the echo can be represented by a linear system of the form Eq. (2.21), determine and plot the impulse response of the echo system Eq. (2.21). Store this impulse response in the vector **he** for $0 \leq n \leq 1000$.

(b). Consider an echo removal system described by the difference equation

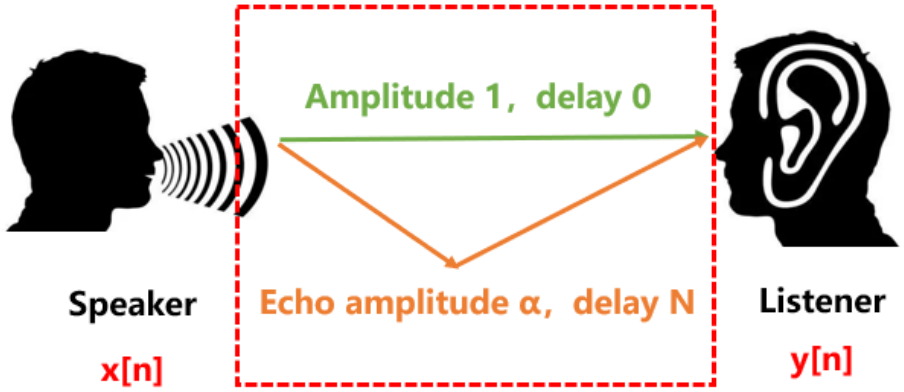
$$z[n] + \alpha z[n - N] = y[n], \tag{2.22}$$

where $y[n]$ is the input and $z[n]$ is the output which has the echo removed. Show that Eq. (2.22) is indeed an inverse of Eq. (2.21) by deriving the overall difference equation relating $z[n]$ to $x[n]$. Is $z[n] = x[n]$ a valid solution to the overall difference equation?

Echo Cancellation: Problem #1

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Problem #1, when only has 1 echo, and α, N given, to get back x[n]?



Voice Channel, as LTI system

y[n]= x[n] + α x[n-N]

$$z[n] + \alpha z[n - N] = y[n],$$

Intermediate Problems

(c). The echo removal system Eq. (2.22) will have an infinite-length impulse response. Assuming that $N = 1000$, and $\alpha = 0.5$, compute the impulse response using `filter` with an input that is an impulse given by `d=[1 zeros(1,4000)]`. Store this 4001 sample approximation to the impulse response in the vector `her`.

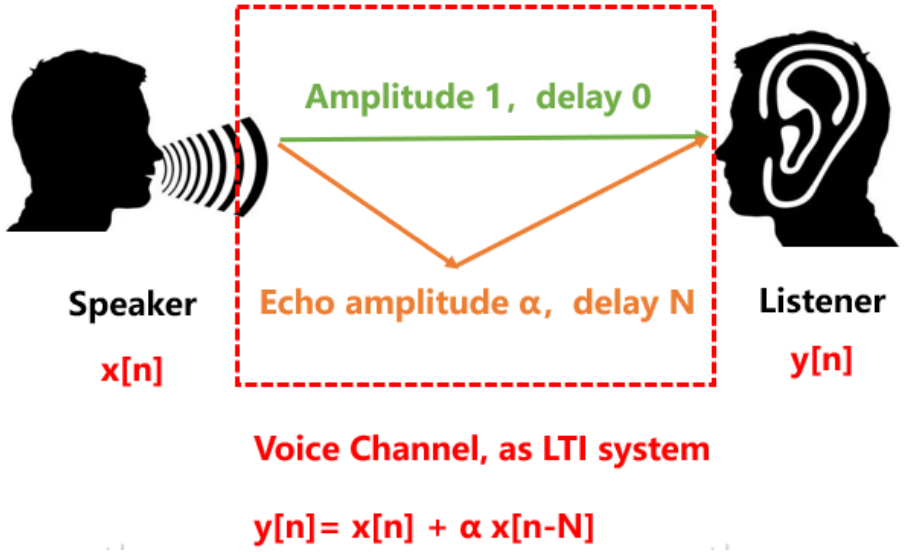
(d). Implement the echo removal system using `z=filter(1,a,y)`, where `a` is the appropriate coefficient vector derived from Eq. (2.22). Plot the output using `plot`. Also, listen to the output using `sound`. You should no longer hear the echo.

(e). Calculate the overall impulse response of the cascaded echo system, Eq. (2.21), and echo removal system, Eq. (2.22), by convolving `he` with `her` and store the result in `hoa`. Plot the overall impulse response. You should notice that the result is not a unit impulse. Given that you have computed `her` to be the inverse of `he`, why is this the case?

Echo Cancellation: Problem #2

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Problem #2, when only has 1 echo, but α, N is unknown, to get back x[n]?



Advanced Problem

(f). Suppose that you were given $y[n]$ but did not know the value of the echo time, N, or the amplitude of the echo, α. Based on Eq. (2.21), can you determine a method of estimating these values? Hint: Consider the output y of the echo system to be of the form:

$$y[n] = x[n] * (\delta[n] + \alpha\delta[n - N])$$

and consider the signal,

$$R_{yy}[n] = y[n] * y[-n].$$

This is called the autocorrelation of the signal $y[n]$ and is often used in applications of echo-time estimation. Write $R_{yy}[n]$ in terms of $R_{xx}[n]$ and also plot $R_{yy}[n]$. You will have to truncate $y[n]$ before your calculations to keep $R_{yy}[n]$ within the limitations

.....

by varying `N`, `alpha`, and `NX`. Also, when you loaded `lineup.mat`, you loaded in two additional vectors. The vector `y2` contains the phrase "line up" with a different echo time N and different echo amplitude α. The vector `y3` contains the same phrase with two echoes, each with different times and amplitudes. Can you estimate N and α for `y2`, and N_1, α_1, N_2, and α_2 for `y3`?

Echo Cancellation：Auto-Correlation

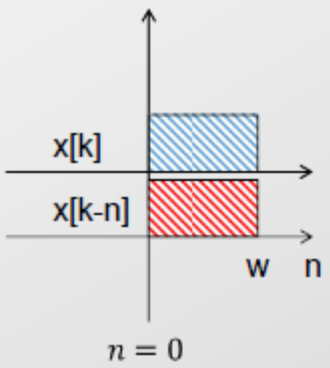
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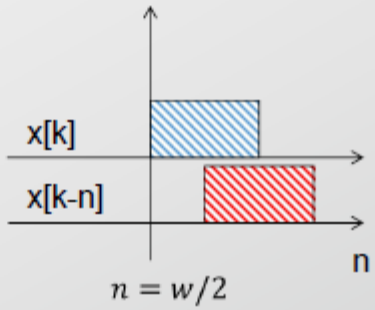
初步认识：自相关定义

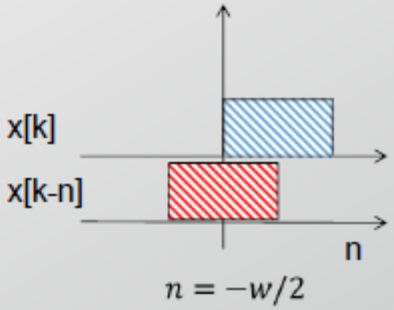
$$R_{xx}(t) = \int_{-\infty}^{+\infty} x(\tau)\, x(\tau - t) d\tau = \int_{-\infty}^{+\infty} x(\tau)\, x(-(t-\tau)) d\tau = x(t) * x(-t)$$

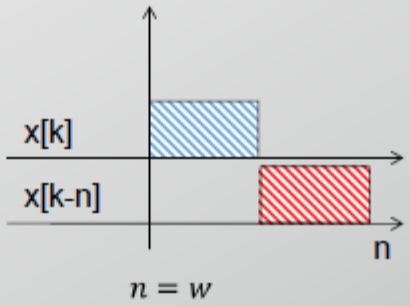
- 用来定量的描述信号 x(t) 与其时移副本 x(t-τ) 的相似程度
- 是信号在时域中特性的平均度量

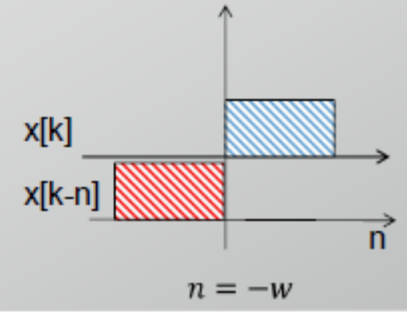
$$Rxx[n] = x[n] * x[-n] = \sum_{k=-\infty}^{+\infty} x[k]x[k-n]$$

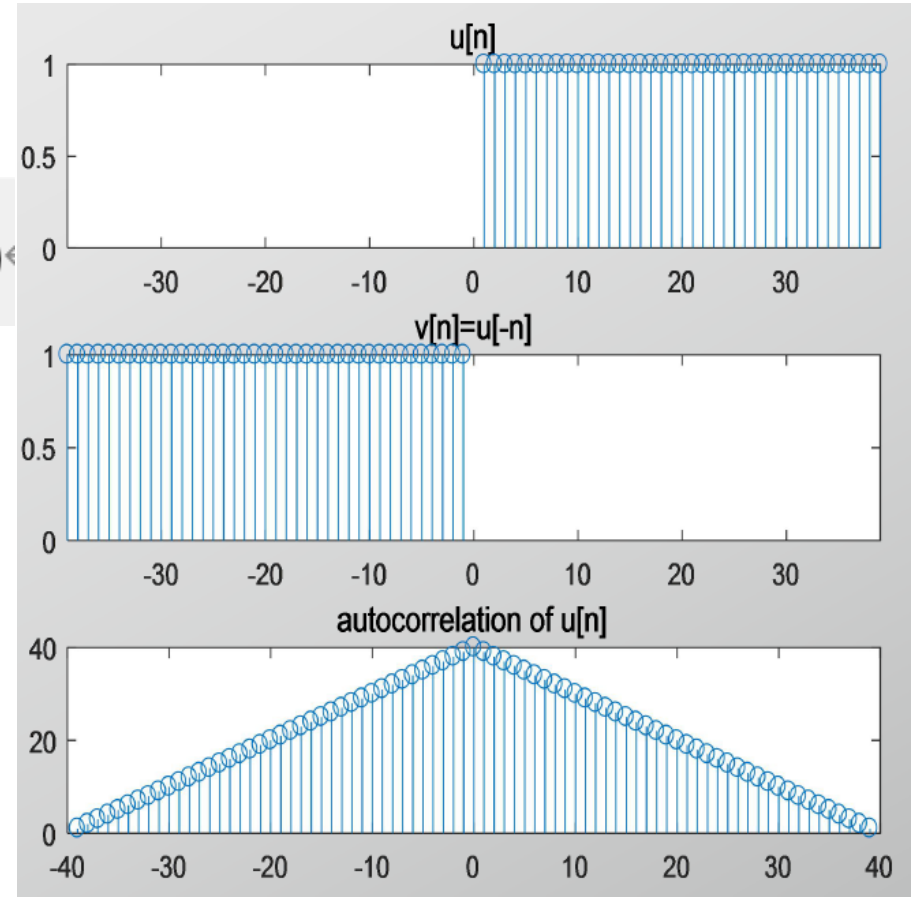












主要性质

- 自相关函数为偶函数，即Y轴对称
- 当 t = 0，自相关函数具有最大值

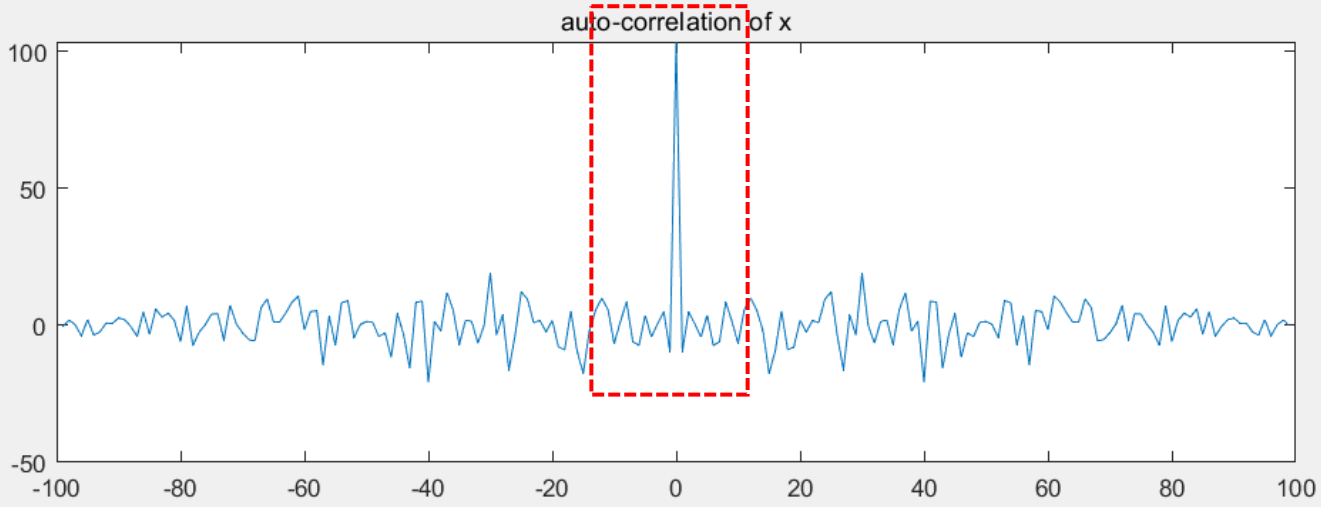
Echo Cancellation：Auto-Correlation

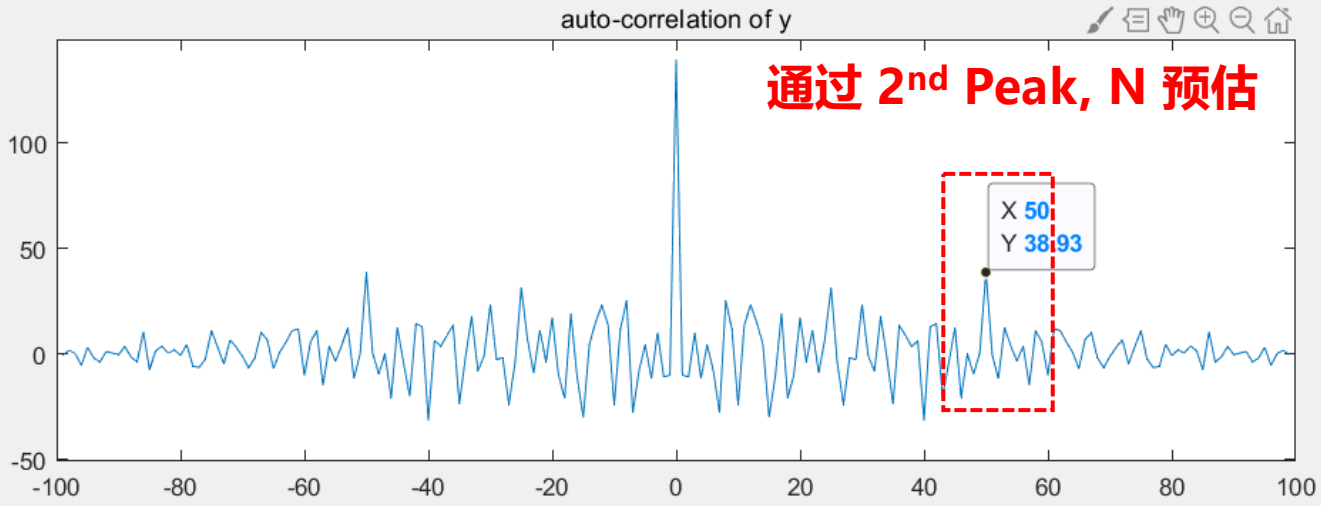
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加深认识：Plot 观察

$$y[n] = x[n] + \alpha x[n - N],$$

```
4       % showcase Rxx, Ryy
5 -     nx = 100;
6 -     x =randn(1, nx);
7
8 -     N = 50;
9 -     alpha = 0.9;
10
11 -    y = filter([1, zeros(1, N-1), alpha], [1], x);
12
13 -    Rxx = conv(x, fliplr(x));
14 -    Ryy = conv(y, fliplr(y));
15
16 -    subplot(2, 1, 1);
17 -    plot([-nx+1 : nx-1], Rxx);
18 -    title("auto-correlation of x");
19
20 -    subplot(2, 1, 2);
21 -    plot([-nx+1 : nx-1], Ryy);
22 -    title("auto-correlation of y");
```





Echo Cancellation: Auto-Correlation

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演算分析

$$Ryy[n] = y[n] * y[-n] = x[n] * (\delta[n] + \alpha\delta[n-N]) * x[-n] * (\delta[-n] + \alpha\delta[-n-N])$$

$$= Rxx[n] * \{(\delta[n] + \alpha\delta[n-N]) * (\delta[-n] + \alpha\delta[-n-N])\}$$

$$= Rxx[n] * (\delta[n] + \alpha\delta[n-N] + \alpha\delta[n+N] + \alpha^2\delta[n-N] * \delta[n+N])$$

$$= Rxx[n] * \big((1+\alpha^2)\delta[n] + \alpha\delta[n+N] + \alpha\delta[n-N]\big)$$

$$= (1+\alpha^2)Rxx[n] + \alpha Rxx[n+N] + \alpha Rxx[n-N]$$

$$Ryy[n] = (1+\alpha^2)Rxx[n] + \alpha\, Rxx[n+N] + \alpha\, Rxx[n-N]$$

$$\Rightarrow \quad Ryy[0] = (1+\alpha^2)Rxx[0] + 2\alpha\, Rxx[N]$$

Note that

$$\delta[n] = \delta[-n]$$

$$\delta[n] * f[n] = f[n]$$

$$\delta[n-N] * f[n] = f[n-N]$$

$$\delta[n-N] * \delta[n+N] = \sum_m \delta[m-N]\,\delta[n-m+N]$$

$$= \sum_m \delta[m-N]\,\delta[m-N-n] = \begin{cases} 1, when = 0 \\ 0, when\ n \neq 0 \end{cases} = \delta[n]$$

Echo Cancellation: Auto-Correlation => N, α

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思路：基于估计N, 并采样Rxx, Ryy的关键点，解方程求解 alpha

$$Ryy[n] = (1+\alpha^2)Rxx[n] + \alpha\, Rxx[n+N] + \alpha\, Rxx[n-N]$$

$$\Rightarrow \quad Ryy[0] = (1+\alpha^2)Rxx[0] + 2\alpha\, Rxx[N]$$

$$Rxx[N] = Rxx[-N]$$

```
41      % find second peak by plot
42 -    N_estimated = 50;
43 -    Rxx_N1 = Rxx(nx - N_estimated);
44 -    Rxx_N2 = Rxx(nx + N_estimated);
45 -    Rxx_0 = Rxx(nx);
46 -    Ryy_0 = Ryy(nx);
47
48 -    syms alpha_s;
49 -    alpha_result = solve((1+alpha_s^2) * Rxx_0 + alpha_s * (Rxx_N1 + Rxx_N2) == Ryy_0, alpha_s);
50 -    alpha_estimated = double(alpha_result);
51 -    disp(alpha_estimated);
```

命令行窗口

```
-0.5319
 0.6330
```

注意到：Matlab 解方程的 API

```
%求解一元二次方程
syms x
result=solve(x^2+2*x-19.25==0,x)
double(result)
```

```
%求解二元二次方程
syms x y;
S=solve([x^2+y*2-17.54==0,x+y==5.8],[x,y])
double(S.x)
double(S.y)
```

Q: 求解的alpha，与 真实alpha 存在较大差异, Why?

Echo Cancellation: Auto-Correlation, Bug??

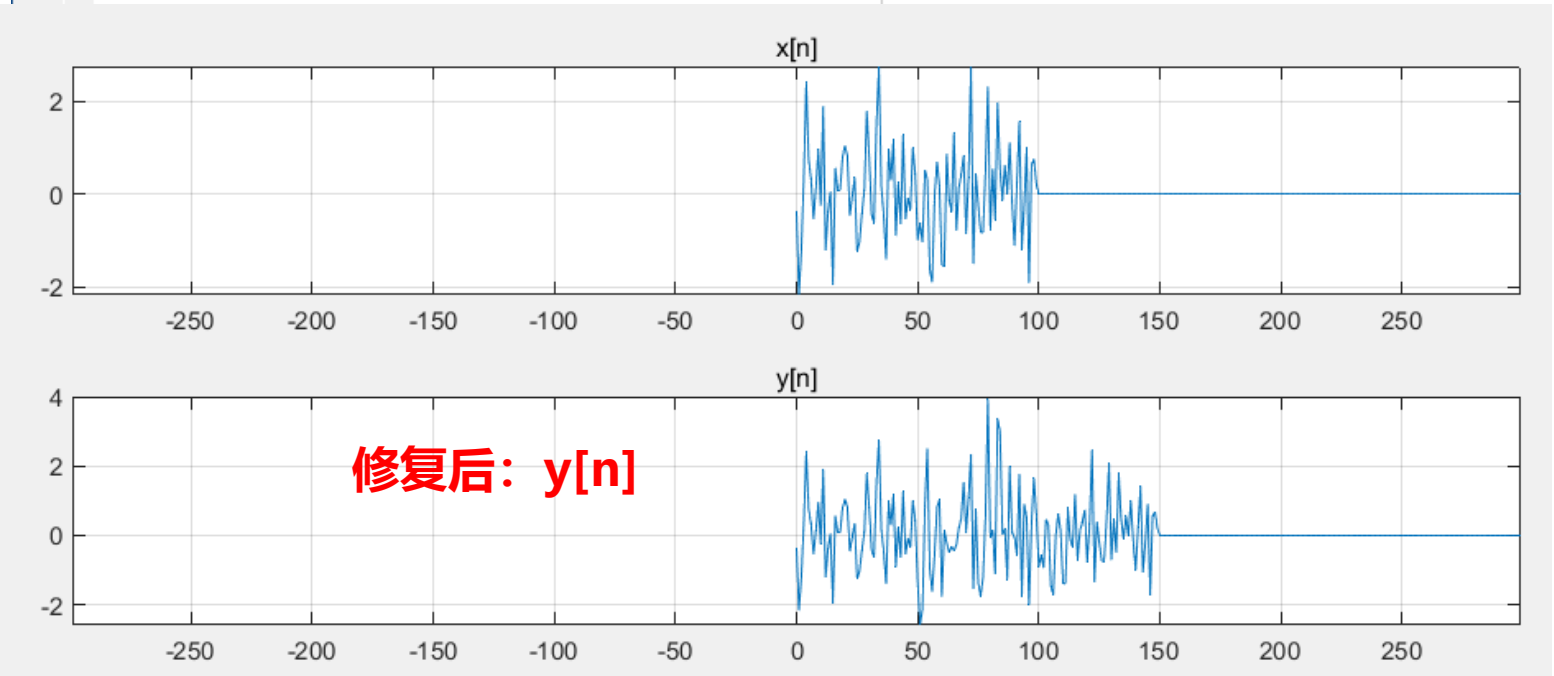
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问题原因：

⇒ Y 被截断了 => Ryy 计算引入偏差

```
4       % showcase Rxx, Ryy
5 -     nx = 100;
6 -     x =randn(1, nx);
7
8 -     N = 50;
9 -     alpha = 0.9;
10
11 -    y = filter([1, zeros(1, N-1), alpha], [1], x);
12
13 -    Rxx = conv(x, fliplr(x));
14 -    Ryy = conv(y, fliplr(y));
15
16 -    subplot(2, 1, 1);
17 -    plot([-nx+1 : nx-1], Rxx);
18 -    title("auto-correlation of x");
19
20 -    subplot(2, 1, 2);
21 -    plot([-nx+1 : nx-1], Ryy);
22 -    title("auto-correlation of y");
```

```
4       % showcase Rxx, Ryy
5 -     nx_nozero = 100;
6 -     nx = 300;
7       % x =randn(1, nx);
8 -     x = [randn(1, nx_nozero), zeros(1, nx-nx_nozero)];
9
10 -    N = 50;
11 -    alpha = 0.9;
12
13 -    y = filter([1, zeros(1, N-1), alpha], [1], x);
```



Echo Cancellation：Auto-Correlation => N, α

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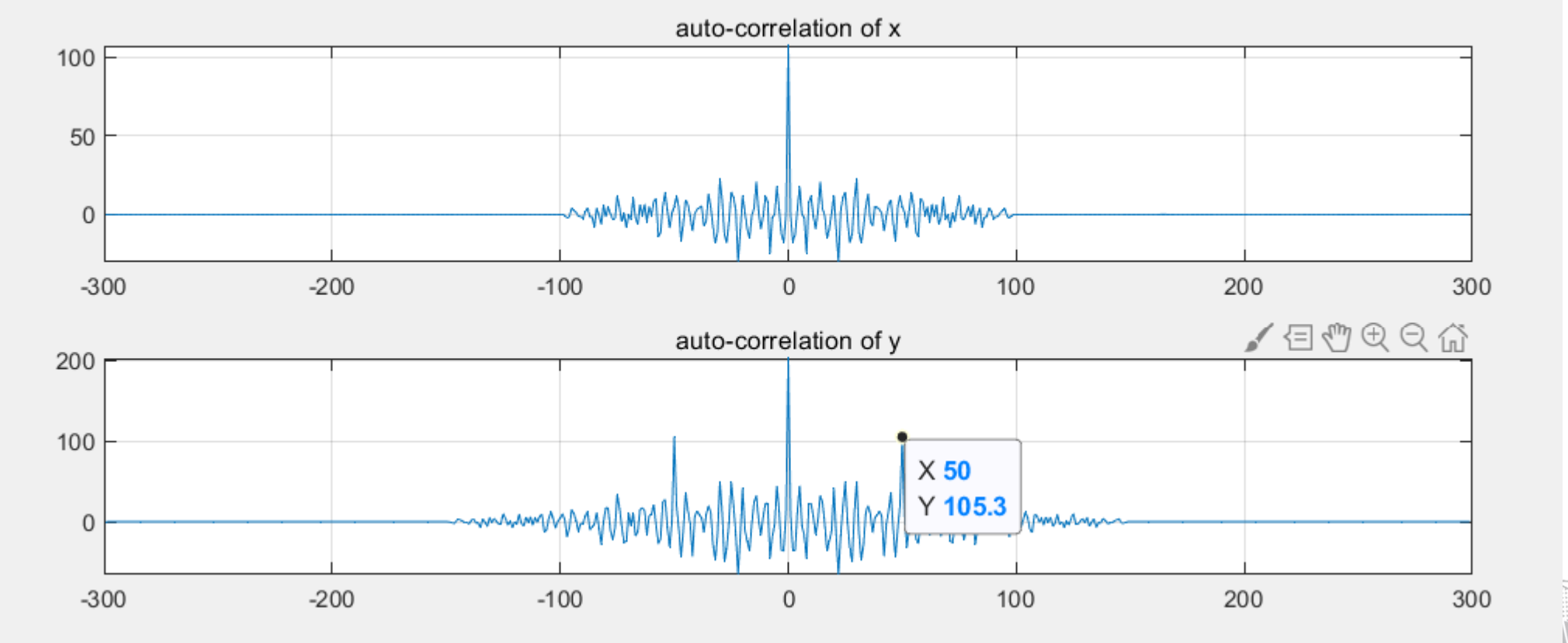
问题解决

- **基于具体情况，选取正的 alpha，与实际吻合**
- **Ryy 的 2nd Peak 更加明显**

命令行窗口

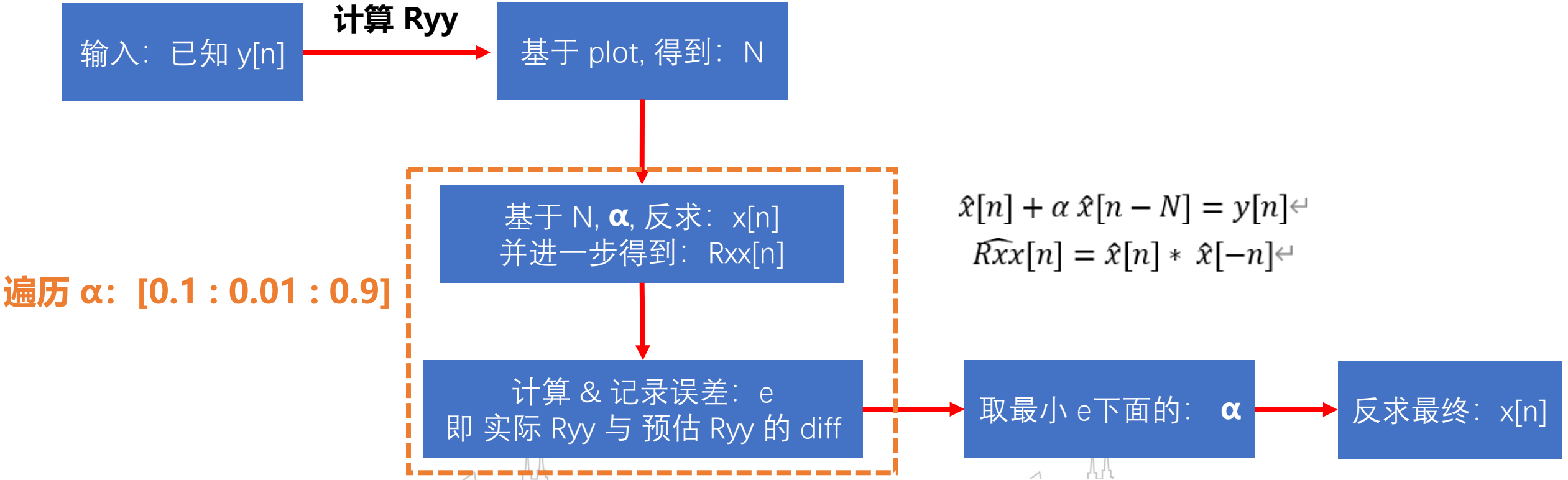
-0.9950

0.9000



Echo Cancellation: Problem #2

Problem #2, when only has 1 echo, but α, N is unknown, to get back x[n]?



$$\hat{x}[n] + \alpha\,\hat{x}[n-N] = y[n]$$

$$\widehat{Rxx}[n] = \hat{x}[n] * \hat{x}[-n]$$

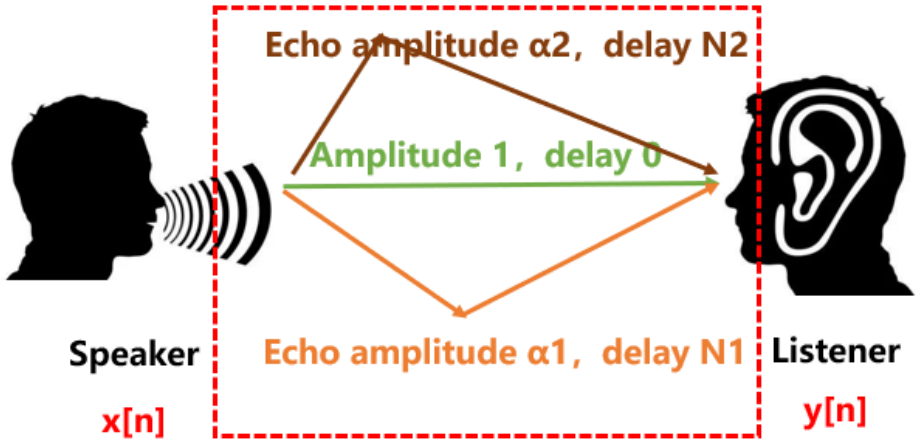
$$\min_{\alpha} \left\| Ryy[n] - \widehat{Ryy}[n] \right\|$$

$$\widehat{Ryy}[n] = (1+\alpha^2)\widehat{Rxx}[n] + \alpha\,\widehat{Rxx}[n+N] + \alpha\,\widehat{Rxx}[n-N]$$

Echo Cancellation: Problem #3

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Problem #3, when has 2 echo, but α1, N1, α2, N2 is unknown, to get back x[n]?



Voice Channel, as LTI system

y[n]= x[n] + α1 x[n-N1] + α2 x[n-N2]

Advanced Problem

(f). Suppose that you were given $y[n]$ but did not know the value of the echo time, N, or the amplitude of the echo, α. Based on Eq. (2.21), can you determine a method of estimating these values? Hint: Consider the output y of the echo system to be of the form:

$$y[n] = x[n] * (\delta[n] + \alpha\delta[n - N])$$

and consider the signal,

$$R_{yy}[n] = y[n] * y[-n].$$

This is called the autocorrelation of the signal $y[n]$ and is often used in applications of echo-time estimation. Write $R_{yy}[n]$ in terms of $R_{xx}[n]$ and also plot $R_{yy}[n]$. You will have to truncate $y[n]$ before your calculations to keep $R_{yy}[n]$ within the limitations

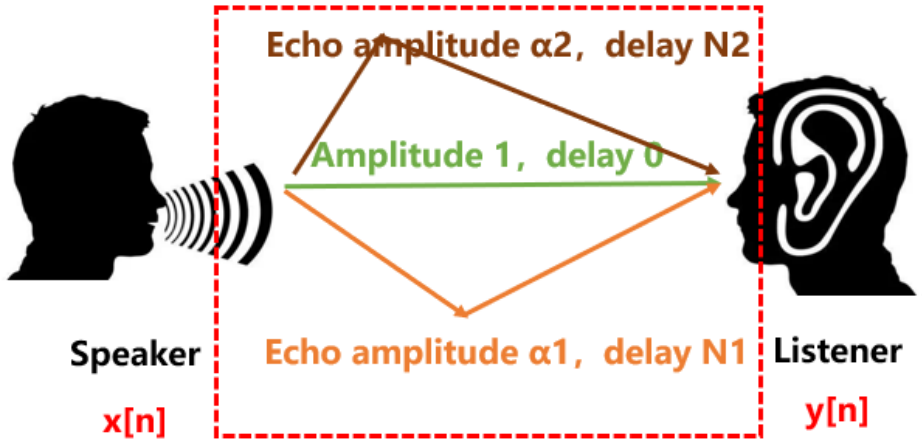
.....

by varying `N`, `alpha`, and `NX`. Also, when you loaded `lineup.mat`, you loaded in two additional vectors. The vector `y2` contains the phrase "line up" with a different echo time N and different echo amplitude α. The vector `y3` contains the same phrase with two echoes, each with different times and amplitudes. Can you estimate N and α for `y2`, and N_1, α_1, N_2, and α_2 for `y3`?

Echo Cancellation: Problem #3

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Problem #3, when has 2 echo, but α1, N1，α2, N2 is unknown, to get back x[n]?



Voice Channel, as LTI system

$y[n] = x[n] + \alpha 1\, x[n-N1] + \alpha 2\, x[n-N2]$

思路提示：

- **step 1), Ryy[n] 推演分析**

$$Ryy[n] = y[n] * y[-n]$$
$$= \{x[n] * (\delta[n] + \alpha_1\delta[n-N_1] + \alpha_2\delta[n-N_2])\}$$
$$* \{x[-n] * (\delta[-n] + \alpha_1\delta[-n-N_1] + \alpha_2\delta[-n-N_2])\}$$
$$= Rxx[n] * ??$$

- **step 2), plot Ryy, 找出 N1, N2 (假设 N1 < N2)**
- **step 3), 遍历 α1, α2 (一般 α1 > α2), 计算 Ryy 预估误差**
- **step 4), 找到误差最小的 α1, α2，反算 x[n]**

LTI System / 课堂练习2

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2.1: d), e)

2.2: g), h), i), j)

2.4: e), f)

2.5:

- **a) ~ c) for system 2**
- **d) ~ g) for system 1**

2.10:

- **a) ~ e), required**
- **f), optional, as bonus**

样本 lineup.mat 下载链接:
https://pan.baidu.com/s/1M3QxJ-gCiVTteWqStO25gA?pwd=r9gp
提取码: r9gp

绿色部分：不用提交，建议练习

红色部分：本次课堂作业

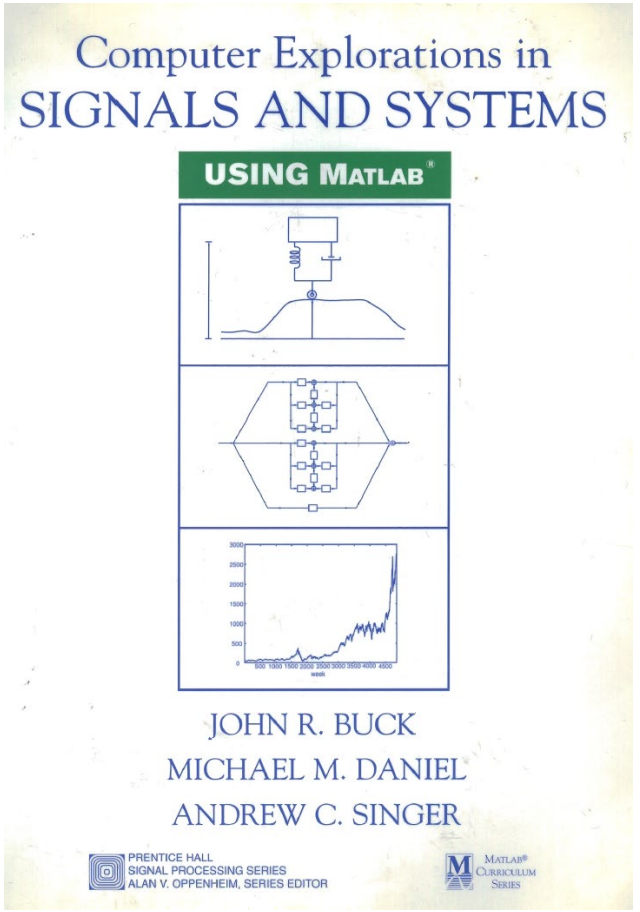
其他：上次课堂作业

Deadline: 5th April, 23:59

内容：2.4, 2.5, 2.10 要求题目

格式: PDF + m代码压缩包

Blackboard：TBD



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THANK YOU

